Asset Allocation

Topics and Applications

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Lecture slides — companion to the book

Course overview

Introduction to Asset Allocation

Portfolio Theory

Robust Covariance Estimation

Backtesting Multi-Asset Strategies

Coding Session 1

Long-Term Expected Returns

Robust Optimization

Views on Markets

Coding Session 2

Macroeconomic and Market Regimes

Tactical Asset Allocation and Dynamic Strategies

Agenda for the Course

General information

1. **Overview:** The objective of this course is to understand the theoretical and practical aspects of asset allocation
2. **Prerequisites:** M1 Finance or equivalent
3. **ECTS:** 3
4. **Keywords:** Finance, Asset Management, Optimization, Statistics
5. **Hours:** Lectures: 18h, HomeWork: 30h
6. **Evaluation:** Project + Class participation
7. **Course website:**

Objective of the course

The objective of the course is twofold:

1. Familiarize students with the subject of asset allocation and how it relates with investing
2. Increase proficiency in quantitative portfolio management

Class schedule

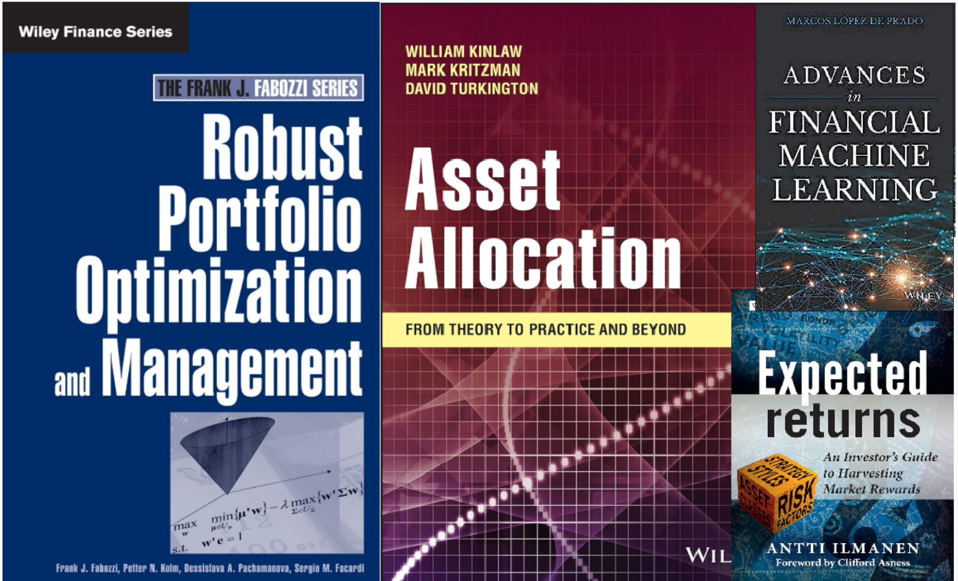
Course sessions

- June 10 (6 hours, 9:30-10:50, 11:00-12:20,14:00-15:20 in room 314)
- June 11 (6 hours, 9:30-10:50, 11:00-12:20,14:00-15:20 in room 314)
- June 12 (4 hours, 9:30-10:50, 11:00-12:20 in room 314)

Requirements

- Class participation (25%)
- Group project (75%)

Recommended books



Introduction to Asset Allocation

Learning objectives

After this lecture you will be able to:

- Define an asset class and distinguish strategic (SAA) from tactical (TAA) allocation
- Place market timing, TAA, and SAA on the investment-horizon spectrum
- Outline the end-to-end investment process and the role of risk management
- Explain why the asset-allocation decision dominates long-run portfolio outcomes

Outline

1. What is an asset class?
2. Asset allocation across horizons: MT, TAA, SAA
3. The investment process
4. How much does asset allocation matter?
5. The efficient frontier and the active manager

What is an asset class?

A group of securities qualifies as an asset class when it is:

1. A stable aggregation, investable
2. Internally homogeneous
3. Externally heterogeneous — distinct from other classes
4. Expected to raise utility when added to the mix
5. Does not necessitate selection skill
6. Accessible cost-effectively

Potential asset classes

- Cash equivalents
- Domestic / foreign Treasury bonds
- Domestic corporate bonds
- Domestic / foreign equities
- Emerging-market equities
- TIPS
- Domestic / foreign real estate
- Commodities
- Infrastructure
- Private equity
- Timber
- Cryptocurrencies / digital assets

A question to keep asking

Which of these are genuine asset classes, and which are not? Or maybe for whom?

Asset allocation across horizons

The same universe is allocated by two decisions on two time scales:

Decision	Horizon	Nature
Market timing (MT)	Intraday – 1 week	Opportunistic
Tactical (TAA)	1 month – 1 year	Active tilts
Strategic (SAA)	5 – 50 years	Policy mix

Key insight

The strategic mix is the slow-moving anchor; TAA are active deviations *around* it, Market timing typically opportunistic rebalancing.

The investment process and governance

Asset allocation sits inside a repeating process run by the investment committee:

1. Investment policy statement (IPS)
2. Strategic asset allocation
3. Tactical asset allocation
4. Securities selection
5. Rebalancing
6. Monitoring
7. Risk management — spanning every step

How much does asset allocation matter?

The classic finding

Brinson et al. (1986) and the follow-up Brinson et al. (1991): SAA explains 70–90% of the *variation* in returns over time; tactical allocation and selection contribute far less.

This number is quoted everywhere — and routinely misread.

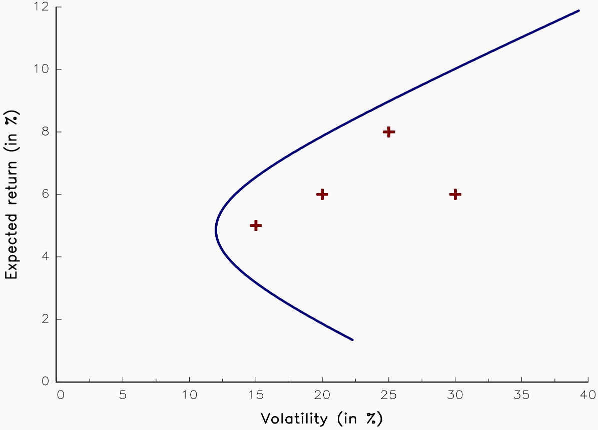
What can go wrong: misreading the 90%

The fallacy

"90% of returns come from asset allocation" is *not* what the studies show.
They explain variation *over time*, implicitly comparing against being uninvested.

- Example: 75% of pfo in tech stocks and 25% in US Bonds. Variance contribution asset allocation 73% vs 27% security selection.
- Against a 60/40 benchmark, the share attributable to the allocation *decision* is much smaller — most variance is just being in the market. Due to asset allocation 16% vs 27% security selection and 57% Default asset mix
- The Samuelson dictum: markets are "micro-efficient, macro-inefficient" — which flips where skill is most likely to pay off

The efficient frontier



Source: Course slide figure pool

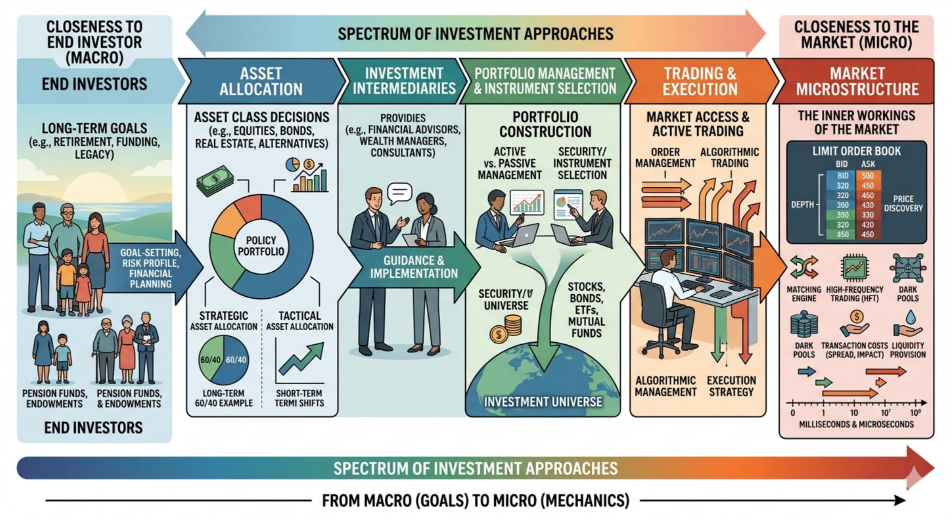
The frontier is the parametric curve

$$\left(\sigma(\boldsymbol{w}^*(\lambda)),\ \mu(\boldsymbol{w}^*(\lambda))\right), \quad \lambda \geq 0,$$

swept out by the risk-aversion parameter λ.

Where can an active manager sit? Only genuine skill pushes a portfolio *above* the frontier built from public information.

Top-down spectrum: where does Asset allocation sit?



Key takeaways

- An asset class is a stable, investable, internally homogeneous group that earns its place by raising utility
- MT, TAA, and SAA differ by horizon; SAA is the long-run anchor
- Risk management runs through the whole investment process, not just the end
- "Asset allocation explains 90%" is about variation over time — read attribution studies carefully

Further reading

Kinlaw et al. (2021) Chapters 1–3 and 10: the asset-allocation decision, from theory to practice.

Fabozzi et al. (2007) Chapters 1 and 14: framing the portfolio problem and its practical management.

Cases and code

Case studies

- Yale investment office
- UBS Risk parity portfolio

The source code for the lectures is available on GitHub: gaa_students GitHub. It contains Jupyter notebooks with the code for all lectures, as well as the datasets used in the course.

Forming groups: 1-2, max group of 3. Groups have to report their member compositions by the beginning of the day 2 and discuss what case study they would like to tackle.

Portfolio Theory

Learning objectives

After this lecture you will be able to:

- Evaluate investments with a utility function and certainty equivalent
- Derive the MVO optimal portfolio and state the two-fund separation theorem
- Compute the tangency portfolio and connect it to CAPM and APT
- Identify where modern portfolio theory fails — statistically and behaviourally

Outline

1. The setup: assets, weights, and portfolio moments
2. Utility and the investor's problem
3. The efficient frontier and two-fund separation
4. Adding a risk-free asset: the tangency portfolio
5. Equilibrium and factors: CAPM, APT, factor investing
6. Worked example, and where the theory fails

Assets and portfolio weights

Consider N risky assets. The investor holds portfolio weights $\boldsymbol{w} = (w_1, \ldots, w_N)^\top$, where w_i is the fraction of wealth in asset i.

Budget constraint (fully invested):

$$\mathbf{1}^\top \boldsymbol{w} = 1 = \sum_{i=1}^{N} w_i$$

Short positions ($w_i < 0$) are permitted for now — we will relax this later.

Inputs:

- $\boldsymbol{\mu} \in \mathbb{R}^N$ — expected returns
- $\boldsymbol{\Sigma} \in \mathbb{R}^{N \times N}$ — covariance matrix

Assumptions:

- $\boldsymbol{\Sigma}$ is symmetric and positive definite
- No redundant assets

Portfolio moments

Portfolio return and risk follow directly from the weights:

Expected return:

$$\mu_p = \mathbb{E}[r_p] = \boldsymbol{w}^\top \boldsymbol{\mu}$$

Variance:

$$\sigma_p^2 = \mathrm{Var}(r_p) = \boldsymbol{w}^\top \Sigma \boldsymbol{w}$$

Key insight

These two equations are the entire scaffolding of the theory.
Everything else is choosing $\boldsymbol{w}$ to balance them.

The Sharpe algorithm: a two-asset example

A mean–variance investor with risk aversion λ_{RA} chooses weights to maximise

$$\mathbb{E}[U] = \mu_p - \lambda_{RA}\,\sigma_p^2, \qquad \sigma_p^2 = w_s^2\sigma_s^2 + w_b^2\sigma_b^2 + 2w_s w_b\,\rho\,\sigma_s\sigma_b.$$

The first-order conditions set each asset's marginal utility to zero:

$$\frac{\partial \mathbb{E}[U]}{\partial w_s} = \mu_s - \lambda_{RA}\left(2w_s\sigma_s^2 + 2w_b\,\rho\,\sigma_s\sigma_b\right) = 0,$$
$$\frac{\partial \mathbb{E}[U]}{\partial w_b} = \mu_b - \lambda_{RA}\left(2w_b\sigma_b^2 + 2w_s\,\rho\,\sigma_s\sigma_b\right) = 0.$$

Key insight

Each condition equates an asset's marginal return μ_i to its marginal risk contribution. Calibrate w_s, w_b until the marginal utility of stocks equals that of bonds.

Further Notations

- $R = (R_1, \ldots, R_N)^\top$ is the vector of asset returns where R_i is the return of asset i
- The return of the portfolio is equal to:

$$R(w) = \sum_{i=1}^{N} w_i R_i = w^\top R$$

- $\mu_p = \mathbb{E}[R]$ and $\Sigma_p = \mathbb{E}\left[(R - \mu_p)(R - \mu_p)^\top\right]$ are the vector of expected returns and the covariance matrix of asset returns

Computation of the first two moments

The expected return of the portfolio is:

$$\mu_p(w) = \mathbb{E}[R(w)] = \mathbb{E}\left[w^\top R\right] = w^\top \mathbb{E}[R] = w^\top \mu$$

whereas its variance is equal to:

$$\begin{aligned}
\sigma_p^2(w) &= \mathbb{E}\left[(R(w) - \mu(w))(R(w) - \mu(w))^\top\right] \\
&= \mathbb{E}\left[\left(w^\top R - w^\top \mu\right)\left(w^\top R - w^\top \mu\right)^\top\right] \\
&= \mathbb{E}\left[w^\top (R - \mu)(R - \mu)^\top w\right] \\
&= w^\top \mathbb{E}\left[(R - \mu)(R - \mu)^\top\right] w \\
&= w^\top \Sigma w
\end{aligned}$$

Utility and investment evaluation

Investors rank uncertain outcomes by expected utility $\mathbb{E}[u(W)]$, not by expected wealth. Concave u ($u'' < 0$) encodes risk aversion.

The certainty equivalent is the certain wealth that matches a gamble:

$$u(\mathrm{CE}) = \mathbb{E}[u(W)], \qquad \mathrm{CE} \leq \mathbb{E}[W]$$

The gap $\mathbb{E}[W] - \mathrm{CE}$ is the risk premium the investor demands. A second-order expansion of this idea is exactly the mean–variance trade-off.

Quadratic utility

A rational investor trades off return against risk. The canonical formalisation uses quadratic utility:

$$U(\boldsymbol{w}) = \underbrace{\boldsymbol{w}^\top \boldsymbol{\mu}}_{\text{return}} - \frac{1}{2}\lambda \underbrace{\boldsymbol{w}^\top \boldsymbol{\Sigma} \boldsymbol{w}}_{\text{variance}}$$

$\lambda > 0$ is the **risk-aversion coefficient** — the exchange rate between return and variance.

Small λ	$\longleftrightarrow$	**Large** λ
Risk tolerant		Risk averse
Aggressive positions		Conservative blend

The MVO solution

Maximising $U(\boldsymbol{w})$ subject to $\mathbf{1}^\top \boldsymbol{w} = 1$ gives a closed-form optimal portfolio:

$$\boldsymbol{w}^* = \frac{1}{\lambda}\boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu} - \gamma\,\mathbf{1}), \qquad \gamma = \frac{\mathbf{1}^\top\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu} - \lambda}{\mathbf{1}^\top\boldsymbol{\Sigma}^{-1}\mathbf{1}}$$

Reading the formula:

- $\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}$ — the *signal-to-noise transform*: tilts weight toward assets with high return per unit of (matrix-weighted) risk
- $\gamma\,\mathbf{1}$ — enforces the budget constraint
- $1/\lambda$ — scales down positions as risk aversion rises

The Lagrangian derivation

Attach a multiplier γ to the budget constraint $\mathbf{1}^\top \boldsymbol{w} = 1$:

$$\mathcal{L}(\boldsymbol{w}, \gamma) = \boldsymbol{w}^\top \boldsymbol{\mu} - \tfrac{1}{2}\lambda\, \boldsymbol{w}^\top \boldsymbol{\Sigma} \boldsymbol{w} - \gamma\big(\mathbf{1}^\top \boldsymbol{w} - 1\big)$$

The first-order conditions set the gradient in $\boldsymbol{w}$ to zero and restate the constraint:

$$\nabla_{\boldsymbol{w}} \mathcal{L} = \boldsymbol{\mu} - \lambda\, \boldsymbol{\Sigma} \boldsymbol{w} - \gamma\, \mathbf{1} = \mathbf{0}, \qquad \mathbf{1}^\top \boldsymbol{w} = 1$$

Solve the first equation for $\boldsymbol{w}$, then substitute into $\mathbf{1}^\top \boldsymbol{w} = 1$ to fix γ:

$$\boldsymbol{w}^* = \tfrac{1}{\lambda}\, \boldsymbol{\Sigma}^{-1}\big(\boldsymbol{\mu} - \gamma\, \mathbf{1}\big), \qquad \gamma = \frac{\mathbf{1}^\top \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu} - \lambda}{\mathbf{1}^\top \boldsymbol{\Sigma}^{-1} \mathbf{1}}$$

Two building blocks: minimum variance + return tilt

Substituting γ back and grouping terms splits the optimum into a λ-free anchor plus a return-seeking tilt:

$$\boldsymbol{w}^*(\lambda) = \underbrace{\frac{\boldsymbol{\Sigma}^{-1}\mathbf{1}}{\mathbf{1}^\top\boldsymbol{\Sigma}^{-1}\mathbf{1}}}_{\text{minimum-variance core}} + \frac{1}{\lambda}\cdot\underbrace{\frac{(\mathbf{1}^\top\boldsymbol{\Sigma}^{-1}\mathbf{1})\,\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu} - (\mathbf{1}^\top\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu})\,\boldsymbol{\Sigma}^{-1}\mathbf{1}}{\mathbf{1}^\top\boldsymbol{\Sigma}^{-1}\mathbf{1}}}_{\text{speculative tilt toward } \boldsymbol{\mu}}$$

Global minimum-variance portfolio

As risk aversion grows without bound the tilt vanishes and only the core survives:

$$\boldsymbol{w}_{\mathrm{mv}} = \boldsymbol{w}^*(\infty) = \frac{\boldsymbol{\Sigma}^{-1}\mathbf{1}}{\mathbf{1}^\top\boldsymbol{\Sigma}^{-1}\mathbf{1}}$$

Quadratic programming problem

Numerical solvers expect a generic quadratic program (here $Q = \Sigma$, $R = \boldsymbol{\mu}$):

$$\boldsymbol{w}^{\star} = \arg\min \ \tfrac{1}{2}\,\boldsymbol{w}^{\top} Q\, \boldsymbol{w} - \boldsymbol{w}^{\top} R$$
$$\text{s.t.} \begin{cases} A\boldsymbol{w} = B \\ C\boldsymbol{w} \leq D \\ \boldsymbol{w}^{-} \leq \boldsymbol{w} \leq \boldsymbol{w}^{+} \end{cases}$$

because the equality and box constraints collapse into a single inequality $S\boldsymbol{w} \leq T$:

$$S\boldsymbol{w} \leq T \ \Leftrightarrow \begin{bmatrix} -A \\ A \\ C \\ -I_n \\ I_n \end{bmatrix} \boldsymbol{w} \leq \begin{bmatrix} -B \\ B \\ D \\ -\boldsymbol{w}^{-} \\ \boldsymbol{w}^{+} \end{bmatrix}$$

Efficient frontier by minimising variance for a target return

For each target return $\bar{\mu}$, find the portfolio with the smallest variance:

$$\min_{\boldsymbol{w}} \ \boldsymbol{w}^\top \boldsymbol{\Sigma} \boldsymbol{w} \quad \text{s.t.} \quad \boldsymbol{w}^\top \boldsymbol{\mu} = \bar{\mu}, \quad \mathbf{1}^\top \boldsymbol{w} = 1$$

Solving for all values of $\bar{\mu}$ traces the **efficient frontier** in (σ_p, μ_p) space.

Key geometric fact: The frontier is a *hyperbola* — the frontier expands as assets become less correlated.

The Sharpe ratio

Suppose the investor can also lend or borrow at a riskless rate r_f. Any portfolio is now $\alpha \boldsymbol{w}_T + (1 - \alpha) r_f$.

Which risky portfolio $\boldsymbol{w}_T$ should all investors hold? The one that maximises the **Sharpe ratio**:

$$S(\boldsymbol{w}) = \frac{\boldsymbol{w}^\top \boldsymbol{\mu} - r_f}{\sqrt{\boldsymbol{w}^\top \boldsymbol{\Sigma} \boldsymbol{w}}} = \frac{\mu_p - r_f}{\sigma_p}$$

Differences in risk aversion affect only α — how much to hold of this portfolio — not *which* portfolio.

The tangency portfolio

The portfolio that maximises $S(\boldsymbol{w})$ has the closed-form solution:

$$\boldsymbol{w}_T = \frac{\Sigma^{-1}(\boldsymbol{\mu} - r_f \mathbf{1})}{\mathbf{1}^\top \Sigma^{-1}(\boldsymbol{\mu} - r_f \mathbf{1})}$$

Key insight

Two-fund separation theorem. Every mean-variance investor holds the same risky portfolio ($\boldsymbol{w}_T$) combined with the risk-free asset (CML). *What to hold* is separated from *how much risk to take*.

$\boldsymbol{w}_T$ is also called the **maximum-Sharpe portfolio** or the **market portfolio** (under CAPM assumptions).

From tangency to CAPM

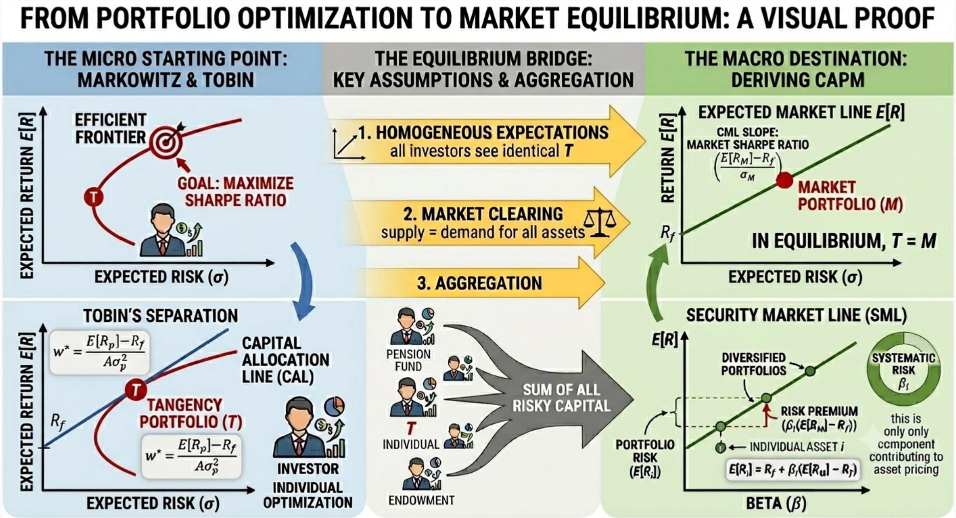
If every investor holds the tangency portfolio, in equilibrium it *is* the market portfolio. Marginal return over risk of adding asset i equals market Sharpe ratio. Expected excess return of each asset i is then linear in market beta (SML):

$$\mathbb{E}[r_i] - r_f = \beta_i\big(\mathbb{E}[r_m] - r_f\big), \qquad \beta_i = \frac{\mathrm{Cov}(r_i, r_m)}{\mathrm{Var}(r_m)}$$

Key insight

Only systematic risk (co-movement with the market) is rewarded. Diversifiable risk earns nothing — you are not paid for what you can diversify away.

CAPM visually



Computation of the beta

The least-squares method

- Let $r_{i,t}$ and $r_{x,t}$ be the returns of asset i and reference portfolio $\boldsymbol{w}_x$ at time t.
- β_i is estimated by the time-series regression

$$r_{i,t} = \alpha_i + \beta_i \, r_{x,t} + \varepsilon_{i,t}$$

- For any portfolio $\boldsymbol{w}_y$ the same regression delivers its beta:

$$r_{y,t} = \alpha_y + \beta_y \, r_{x,t} + \varepsilon_{y,t}$$

Computation of the beta

The covariance method

Equivalently, the beta of portfolio $\boldsymbol{w}_y$ relative to $\boldsymbol{w}_x$ is their covariance over the variance of $\boldsymbol{w}_x$:

$$\beta(\boldsymbol{w}_y \mid \boldsymbol{w}_x) = \frac{\mathrm{Cov}(r_y, r_x)}{\mathrm{Var}(r_x)} = \frac{\boldsymbol{w}_y^\top \boldsymbol{\Sigma}\, \boldsymbol{w}_x}{\boldsymbol{w}_x^\top \boldsymbol{\Sigma}\, \boldsymbol{w}_x}$$

Setting $\boldsymbol{w}_y = \boldsymbol{e}_i$ (a unit holding in asset i) recovers the asset beta:

$$\beta_i = \beta(\boldsymbol{e}_i \mid \boldsymbol{w}_x) = \frac{\boldsymbol{e}_i^\top \boldsymbol{\Sigma}\, \boldsymbol{w}_x}{\boldsymbol{w}_x^\top \boldsymbol{\Sigma}\, \boldsymbol{w}_x} = \frac{(\boldsymbol{\Sigma}\, \boldsymbol{w}_x)_i}{\boldsymbol{w}_x^\top \boldsymbol{\Sigma}\, \boldsymbol{w}_x}$$

A portfolio's beta is the weighted average of its constituents' betas:

$$\beta(\boldsymbol{w}_y \mid \boldsymbol{w}_x) = \boldsymbol{w}_y^\top \frac{\boldsymbol{\Sigma}\, \boldsymbol{w}_x}{\boldsymbol{w}_x^\top \boldsymbol{\Sigma}\, \boldsymbol{w}_x} = \sum_{i=1}^{N} w_{y,i}\, \beta_i$$

Why one period is not enough

The CAPM is static: investors care only about end-of-period wealth and treat the investment opportunity set — r_f, expected returns, volatilities — as fixed.

In reality those quantities are stochastic state variables that drift over time. A long-horizon investor cares not just about wealth today, but about the terms on which that wealth can be reinvested tomorrow.

Key insight

When the opportunity set shifts, multi-period investors take extra positions to hedge adverse shifts — e.g. assets that pay off when r_f falls. This hedging demand sits on top of the myopic mean–variance demand.

Merton's intertemporal CAPM

Merton (1973) solves the multi-period problem in continuous time. Equilibrium expected excess returns load on the market *and* on K state variables $s_1, \ldots, s_K$ that describe the changing opportunity set:

$$\mathbb{E}[r_i] - r_f = \beta_{i,m}\big(\mathbb{E}[r_m] - r_f\big) + \sum_{k=1}^{K} \beta_{i,k}\,\lambda_k$$

- $\beta_{i,m}$ — the familiar market exposure (the myopic term)
- $\beta_{i,k}$ — exposure to innovations in state variable s_k; λ_k is its hedging risk premium
- Set $K = 0$ (constant opportunity set) and the ICAPM collapses back to the CAPM

Key insight

The ICAPM gives economic content to the extra factors that APT will price by

Arbitrage pricing theory

The general asset-pricing model sets $\mathbb{E}[r_i] = r_f + g(f_1, \ldots, f_K)$. Ross (1976) specialises g to be linear in K common factors:

$$r_i = \alpha_i + \sum_{k=1}^{K} \beta_{ik} f_k + \varepsilon_i \quad \Longleftrightarrow \quad \boldsymbol{r} = \boldsymbol{\alpha} + \boldsymbol{B}\boldsymbol{f} + \boldsymbol{\varepsilon}, \quad \boldsymbol{B} \in \mathbb{R}^{N \times K}$$

Assumptions (often left implicit)

- Zero-mean factors and residuals: $\mathbb{E}[f_k] = \mathbb{E}[\varepsilon_i] = 0$
- Idiosyncratic, homoscedastic residuals: $\mathbb{E}[\varepsilon_i \varepsilon_j] = \mathbb{E}[\varepsilon_i f_k] = 0$ $(i \neq j)$, $\mathbb{E}[\varepsilon_i^2] = \sigma^2$
- Mutually uncorrelated factors: $\mathbb{E}[f_k f_l] = 0$ $(k \neq l)$
- Many assets (N large), so idiosyncratic risk diversifies away

Arbitrage pricing theory: the pricing relation

No arbitrage — no risk-free payoff for zero net investment — forces expected returns to be linear in the factor exposures:

$$\mathbb{E}[r_i] = r_f + \sum_{k=1}^{K} \beta_{ik}\,\lambda_k$$

- λ_k is the risk premium on factor k; for a traded excess-return factor, $\lambda_k = \mathbb{E}[f_k] - r_f$
- Only the exposures β_{ik} are priced — idiosyncratic risk ε_i earns nothing
- CAPM is the $K = 1$ case, with the market as the single factor
- Caveat: APT fixes the *form* of the relation, not the factors themselves — and holds only *approximately* at finite N

Factor investing

APT is agnostic about *which* factors matter; empirical work fills them in:

- Market, size, value, momentum, quality, low-volatility, carry
- Each is a long/short portfolio harvesting a persistent risk premium
- Asset allocation is increasingly framed as allocating to factors, not just asset classes

Ang (2014): "assets are bundles of factor exposures."

Different theories, same factor structure

The Landscape of Asset Pricing Models

How Modern Multi-Factor Models Extend and Alternative to Classical CAPM

Capital Asset Pricing Model (CAPM)

$$E[R_i] = R_f + \beta_{i,M}(E[R_M] - R_f)$$

- Perspective: Single-period utility optimizer
- Key Driver: Aggregate wealth (Market Portfolio)
- Risk Concept: Mean-Variance efficiency
- Core Assumption: Homogeneous expectations

Theoretical Extension (Add Time & Horizon)

Empirical Alternative (Bypass Market Optimization)

Merton's Intertemporal CAPM (ICAPM)

$$E[R_i] = R_f + \beta_{i,M}\lambda_M + \sum_{k=1}^{K} \beta_{i,X_k}\lambda_k$$

- Perspective: Multi-period lifetime utility optimization
- Key Driver: Shifting investment opportunity sets over time
- Risk Concept: Market risk + Hedges against state changes
- State Variables: Macro shocks (Interest rates, inflation, volatility)
- Connection: Collapses to CAPM if opportunities are static

Arbitrage Pricing Theory (APT)

$$E[R_i] = R_f + \beta_{i,1}\lambda_1 + \beta_{i,2}\lambda_2 + \ldots + \beta_{i,K}\lambda_K$$

- Perspective: Macro/No-Arbitrage (Law of One Price)
- Key Driver: Well-diversified portfolios eliminate idiosyncratic risk
- Risk Concept: Multi-factor systematic linear risk exposure
- Factor Source: Empirically driven (PCA, Fama-French, Macro data)
- Connection: Bypasses utility functions and the Market Portfolio

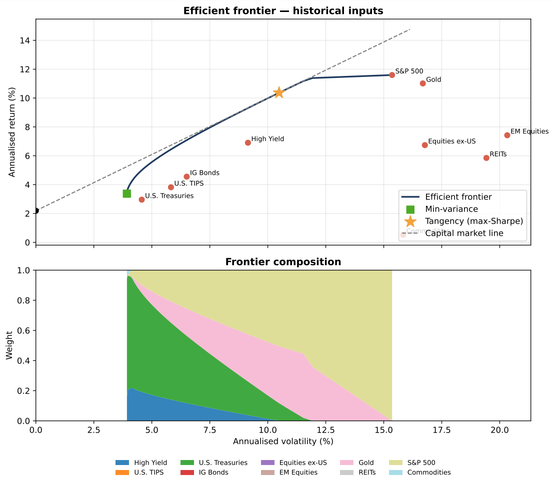
**Note: β denotes sensitivity to a specific risk factor, and λ represents the risk premium associated with that factor.*

A global multi-asset universe

Monthly total returns, 2006–2025 (annualised). Money market is the risk-free benchmark; inputs come straight from `analytics.timeseries_analyzer`.

Asset class	Return (%)	Volatility (%)	Sharpe
S&P 500	11.6	15.4	0.61
Gold	11.0	16.7	0.53
EM equities	7.4	20.3	0.26
High yield	6.9	9.1	0.51
Equities ex-US	6.7	16.8	0.27
REITs	5.8	19.4	0.19
IG bonds	4.6	6.5	0.36
U.S. TIPS	3.8	5.8	0.28
U.S. Treasuries	3.0	4.6	0.17
Commodities	0.5	15.8	−0.11

The efficient frontier and its composition



Source: *book/slides/notebooks/AssetAllocation2_Optimization.ipynb*

Optimal portfolios by objective

Same μ and sample Σ, six objectives solved with `optimization.mean_risk` (long-only):

Objective	Return (%)	Volatility (%)	Sharpe
Min variance	3.4	3.9	0.30
Target return 6%	6.0	5.4	0.71
Max utility ($\lambda = 10$)	8.6	8.2	0.77
Target vol 10%	10.0	10.0	0.78
Max Sharpe	10.4	10.5	0.78
Max utility ($\lambda = 2$)	11.4	11.8	0.78

Notice: the minimum-variance portfolio earns a low Sharpe (0.30); the tangency lifts it to 0.78, and the aggressive objectives concentrate in equities and gold.

The error maximizer in numbers

Drop the long-only constraint and re-solve for the tangency portfolio:

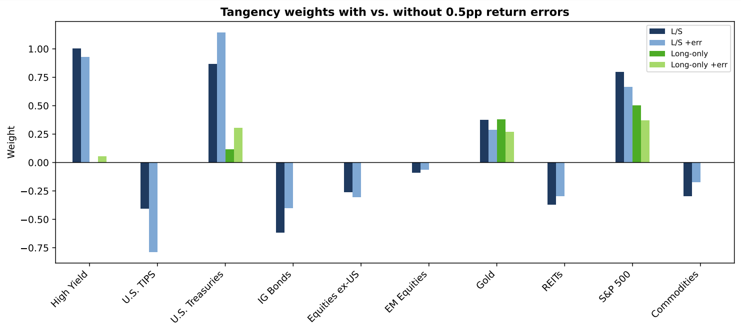
Asset	L/S (%)	Long-only (%)
High yield	100	0
U.S. Treasuries	87	12
S&P 500	80	51
IG bonds	−62	0

Offsetting bets on collinear assets

Unconstrained, the optimizer takes a −62% IG-bond / +87% Treasury spread — gross exposure 509% vs 100% — to harvest an illusory in-sample Sharpe edge (1.39 vs 0.78).

Tiny input errors, large weight swings

Add random 0.5pp errors to μ and re-solve the tangency portfolio:



Long/short (blues): the book lurches — TIPS to -79%, Treasuries to $+114\%$ (turnover ≈ 1.4).

Long-only (greens): barely moves (turnover ≈ 0.5) — the constraint absorbs the noise.

Error maximization in action

The optimizer does exactly what we asked

It loads aggressively on directions in (μ, Σ)-space with the highest apparent signal-to-noise ratio.

Whether those directions reflect...

- Genuine return premia
- Or sampling error in the inputs

...the optimizer cannot tell.

A common fix: Long-only constraint $w_i \geq 0$.

- Rules out short positions
- Dramatically stabilises the solution
- Acts as an implicit form of regularisation

The fundamental problem

Inputs are estimated, not known

μ and Σ must be estimated from finite samples. MVO is exquisitely sensitive to those estimates.

Michaud (1989) called the result an **error maximizer**: the optimizer amplifies whatever errors happen to be in the inputs.

This is not a flaw in the mathematics — it is a consequence of asking a powerful optimiser to solve a noisy estimation problem.

Three failure modes

1. **Expected returns dominate.** Small changes in $\boldsymbol{\mu}$ produce large changes in $\boldsymbol{w}^*$. $\boldsymbol{\mu}$ is the hardest quantity in finance to estimate.

2. **Near-singular covariance matrices.** When N is large relative to the sample size T, $\boldsymbol{\Sigma}^{-1}$ amplifies estimation error. Ledoit and Wolf (2003) showed shrinkage estimators substantially mitigate this.

3. **Concentration and corner solutions.** Unconstrained MVO concentrates in a few assets with large offsetting positions, magnifying transaction costs.

The behavioural critique

Expected-utility theory assumes investors care about *final wealth*. Kahneman and Tversky (1979) show they care about gains and losses relative to a reference point:

- Loss aversion — losses hurt about twice as much as equal gains please
- Probability weighting — small probabilities are over-weighted
- The value function is concave over gains, convex over losses
- Humans engage in mental accounting

Barberis et al. (2021) link these features to asset-pricing anomalies — a different lens on "where the theory fails."

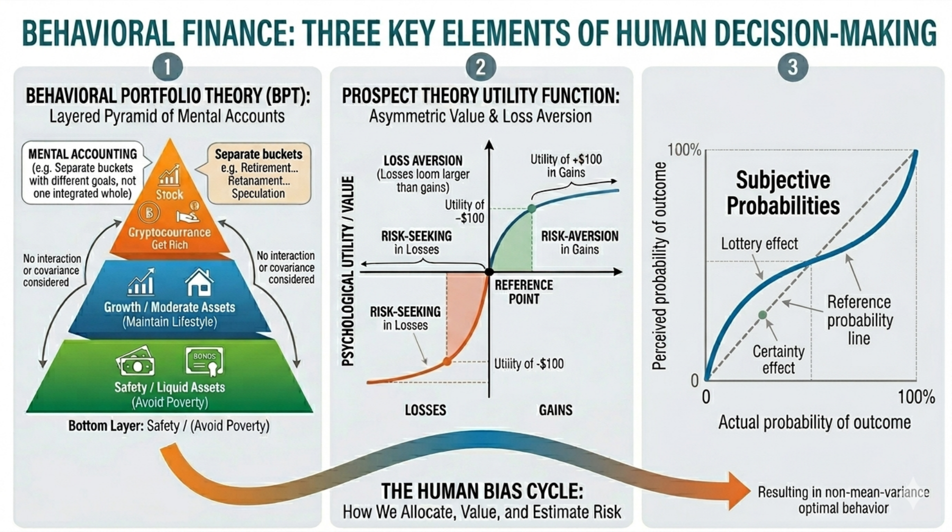
The prospect-theory value function

Replace utility over final wealth $u(W)$ with a value function over gains and losses relative to a reference point θ (last period's wealth, or a benchmark):

$$v(x) = \begin{cases} x^{\alpha}, & x \geq 0 \quad \text{(gain)} \\ -\lambda\,(-x)^{\beta}, & x < 0 \quad \text{(loss)} \end{cases} \qquad x = r_p - \theta$$

Tversky and Kahneman (1992) estimate $\alpha = \beta \approx 0.88$ (diminishing sensitivity) and $\lambda \approx 2.25$ — losses loom about $2.25\times$ larger than equal gains. Outcomes are then weighted by a probability-weighting function $w(p)$ that over-weights the tails.

Behavioural building blocks



Full-scale optimisation

MVO compresses the distribution into $(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and assumes quadratic utility. Full-scale optimisation drops both: score each candidate portfolio by the average value it would have delivered across the *entire* return history.

$$\widehat{U}(\boldsymbol{w}) = \frac{1}{T}\sum_{t=1}^{T} v\big(\boldsymbol{w}^\top \boldsymbol{r}_t - \theta\big)$$

Grid search

- Enumerate $\boldsymbol{w} \in \mathcal{W}$ on a discrete grid (e.g. 5% steps, $\sum_i w_i = 1$)
- Re-score $\widehat{U}(\boldsymbol{w})$ on every historical period

Choose the best

- Pick $\boldsymbol{w}^\star = \arg\max_{\boldsymbol{w}} \widehat{U}(\boldsymbol{w})$
- Check stability across sub-samples before trusting it

No closed form — the value function is kinked and non-quadratic — but it sees

Risk aversion and risk measures for realistic investors

What is risk aversion, really? Not a single λ over wealth. Investors are loss-averse around a reference point ($\lambda \approx 2$), nearly insensitive to small gains, and often risk-seeking over losses (the disposition effect).

Which risk measures make sense? Variance penalises upside and downside symmetrically — realistic investors fear only the downside:

- **Semivariance / downside deviation** — dispersion of returns below the target
- **VaR** — a quantile loss threshold (not coherent; ignores tail shape)
- **CVaR / expected shortfall** — mean loss beyond VaR; coherent and convex (Rockafellar and Uryasev, 2002)
- **Maximum drawdown** — the loss path investors actually live through

Partial remedies (preview)

Failure mode	Remedy (later lectures)
Sensitive to $\boldsymbol{\mu}$	Black–Litterman (Lecture 8), shrinkage
Near-singular $\boldsymbol{\Sigma}$	Ledoit–Wolf shrinkage (Lecture 3), factor models
Concentration / short positions	Long-only constraints, risk-based methods
All three simultaneously	Robust optimisation (Lecture 7)

Most of the rest of the course is, in one way or another, about these remedies.

Key takeaways

- Investors rank gambles by expected utility; the certainty equivalent prices risk
- MVO formalises the return–risk trade-off via quadratic utility; λ is the rate of substitution between risk and return
- The frontier is spanned by two funds (**two-fund separation**); all holding combinations of a risk-free asset and the **tangency portfolio** (**two-funds**)
- In equilibrium the tangency portfolio is the market: CAPM, generalised by APT to many priced **factors**
- MVO is an **error maximizer**; prospect theory adds a behavioural critique of the whole expected-utility framework

Further reading

Kinlaw et al. (2021) Chapters 7, 8, 12: error maximization, factors, MVO.

Fabozzi et al. (2007) Chapters 2–5, 9: portfolio theory.

Markowitz (1952) The original 14-page paper; the insight is clearest in the source.

Sharpe (1964) The CAPM: equilibrium pricing of systematic risk.

Merton (1973) The intertemporal CAPM: hedging demands and state-variable factors.

Ross (1976) Arbitrage pricing theory and multi-factor returns.

Ang (2014) Factor investing: assets as bundles of factor exposures.

Kahneman and Tversky (1979) Prospect theory: the behavioural alternative to expected utility.

Robust Covariance Estimation

Learning objectives

After this lecture you will be able to:

- State the sample covariance estimator and explain why it degrades when N is large relative to T
- Apply EWMA and Ledoit–Wolf shrinkage to stabilise the covariance matrix
- Use a factor model to impose structure and denoise Σ
- Correct the smoothed returns of illiquid assets before estimating risk

Outline

1. The sample covariance matrix and its pathologies
2. Denoising: factor structure, RMT, shrinkage
3. Ledoit–Wolf shrinkage
4. Dynamic models: EWMA
5. Illiquid assets: unsmoothing returns
6. Constructing minimum-variance portfolios

The sample covariance matrix

The textbook estimator from T observations of an N-vector of returns:

$$\widehat{\Sigma} = \frac{1}{T}\sum_{t=1}^{T}\left(R_t - \bar{R}\right)\left(R_t - \bar{R}\right)^{\top}$$

It is unbiased — but it has $N(N+1)/2$ free parameters.

Key insight

The number of parameters grows *quadratically* in N. When N approaches T, $\widehat{\Sigma}$ is noisy and nearly singular, and $\widehat{\Sigma}^{-1}$ amplifies that noise.

Markowitz's curse: a two-asset example

Two unit-variance assets, correlation ρ (López de Prado, 2016). Determinant and condition number $\kappa = \lambda_{\text{max}}/\lambda_{\text{min}}$ in closed form:

$$\boldsymbol{\Sigma} = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}, \qquad \det \boldsymbol{\Sigma} = 1 - \rho^2, \qquad \kappa = \frac{1 + |\rho|}{1 - |\rho|}$$

ρ	$\det \boldsymbol{\Sigma}$	κ
0.0	1.00	1
0.5	0.75	3
0.9	0.19	19
0.99	0.02	199

Key insight

As $\rho \to 1$: $\kappa \to \infty$ and $\boldsymbol{\Sigma}^{-1} \sim 1/(1 - \rho^2)$ explodes. Markowitz's curse: more correlation, less stable weights. *RMT:* noise fills the Marcenko–Pastur band $(1 \pm \sqrt{N/T})^2$, so κ diverges as $N/T \to 1$.

Ways to denoise the covariance matrix

1. Factor analysis — impose a correlation structure (e.g. Barra)
2. Factor filtering — APT-style filtering of the structure
3. Principal component analysis
4. Random matrix theory (RMT)
5. Shrinkage methods

All trade a little bias for a large reduction in estimation variance. We focus on factor structure and shrinkage.

Factor covariance

Assume a linear factor model for returns, with $\mathbf{B}$ the $N \times K$ loadings:

$$\mathbf{r} = \boldsymbol{\mu} + \mathbf{B}\mathbf{f} + \boldsymbol{\varepsilon}$$

If residuals are independent of factors, the covariance matrix collapses to a low-rank-plus-diagonal form:

$$\boldsymbol{\Sigma} = \mathbf{B}\,\mathbf{F}\,\mathbf{B}^{\top} + \mathbf{D}$$

where $\mathbf{F}$ is the $K \times K$ factor covariance and $\mathbf{D}$ the diagonal of specific variances. $K \ll N$ parameters, overall $K(K+1)/2 + \mathrm{N} \times \mathrm{K}$ params instead of $N(N+1)/2$.

Ledoit–Wolf shrinkage

Combine the noisy-but-unbiased $\widehat{\boldsymbol{\Sigma}}$ with a biased-but-stable target $\widehat{\Phi}$ (e.g. constant correlation):

$$\widehat{\boldsymbol{\Sigma}}_\alpha = \alpha\,\widehat{\Phi} + (1-\alpha)\,\widehat{\boldsymbol{\Sigma}}$$

The optimal intensity minimises expected quadratic loss:

$$\alpha^\star = \arg\min_\alpha \mathbb{E}\big[\,\|\alpha\widehat{\Phi} + (1-\alpha)\widehat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}\|^2\,\big]$$

Ledoit and Wolf (2004) derive $\alpha^\star$ in closed form.

Target I: single-factor (market) model

The simplest target ties every covariance to one common factor — the market. Regress each asset on the market return to get its beta β_i:

$$\widehat{\Phi}_{ij} = \beta_i \beta_j \,\widehat{\sigma}^2_m \quad (i \neq j), \qquad \widehat{\Phi}_{ii} = \widehat{\sigma}_{ii}$$

Equivalently $\widehat{\Phi} = \widehat{\sigma}^2_m \boldsymbol{\beta}\boldsymbol{\beta}^\top + \mathbf{D}$, with $\widehat{\sigma}^2_m$ the market variance and $\mathbf{D}$ diagonal so the variances match the sample.

Key insight

Only $N + 1$ parameters: every off-diagonal is pinned down by two betas.

Target II: constant correlation

Keep the sample variances, but replace every pairwise correlation by the single universe-average correlation $\bar{\rho}$:

$$\widehat{\Phi}_{ii} = \widehat{\sigma}_{ii}, \qquad \widehat{\Phi}_{ij} = \bar{\rho}\sqrt{\widehat{\sigma}_{ii}\,\widehat{\sigma}_{jj}}$$

where $\bar{\rho} = \dfrac{2}{N(N-1)} \displaystyle\sum_{i<j} \dfrac{\widehat{\sigma}_{ij}}{\sqrt{\widehat{\sigma}_{ii}\,\widehat{\sigma}_{jj}}}$ averages the sample correlations.

Key insight

The default Ledoit–Wolf target: N variances $+\,1$ correlation. Ideal for a *single* asset class.

Target III: multi-factor

With K common factors the target is the low-rank-plus-diagonal factor covariance from before:

$$\widehat{\Phi} = \mathbf{B}\,\widehat{\mathbf{F}}\,\mathbf{B}^\top + \mathbf{D}$$

$\mathbf{B}$ ($N \times K$ loadings) and $\widehat{\mathbf{F}}$ ($K \times K$ factor covariance) come from a time-series regression on the factors; $\mathbf{D}$ holds the residual variances.

Key insight

Richer structure for *multi-asset-class* universes, where one average correlation is too crude and the constant-correlation target shrinks little.

How much to shrink?

Eight equity indices, weekly, 2005–2011

For a single-asset-class equity universe, the estimated intensity is $\alpha^\star \approx 51\%$ — shrinkage matters a lot.

But across asset classes

For a *multi-asset-class* universe the optimal intensity can be lower $\alpha^\star \approx 20\% - 40\%$: the constant-correlation target is a poor description when blocks of assets behave very differently, but single or multi-factor models will do fine.

The target must match the structure of the universe.

EWMA covariance

For short-horizon (TAA) risk, weight recent observations more heavily:

$$\widehat{\boldsymbol{\Sigma}}_t = \lambda\, \widehat{\boldsymbol{\Sigma}}_{t-1} + (1-\lambda)\, R_{t-1} R_{t-1}^\top$$

- RiskMetrics memory coefficient: $\lambda = 0.94$
- Dynamic, trivial to implement, guaranteed positive semi-definite
- Shrinkage or a factor model can still be layered on top

GARCH/DCC models refine this further but scale poorly in N.

Unsmoothing illiquid returns

Private equity and appraisal-priced assets report serially correlated, artificially smooth returns — understating volatility and correlation.

Geltner's model: the reported return blends the true return with its own lag,

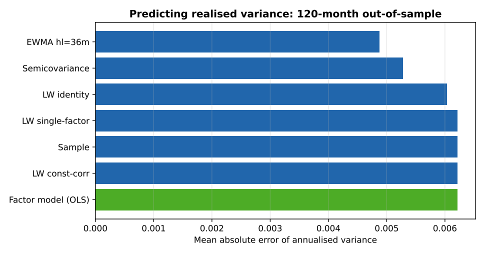
$$r_t = (1 - c)\, r_t^\star + c\, r_{t-1}$$

Inverting the filter recovers the economic return:

$$r_t^\star = \frac{r_t - c\, r_{t-1}}{1 - c}, \qquad c = \rho_1 \text{ (first-order autocorrelation)}$$

Predicting future variance

Score each model's predicted variances against the realised variances of the next 120 months (mean absolute error across 11 assets).

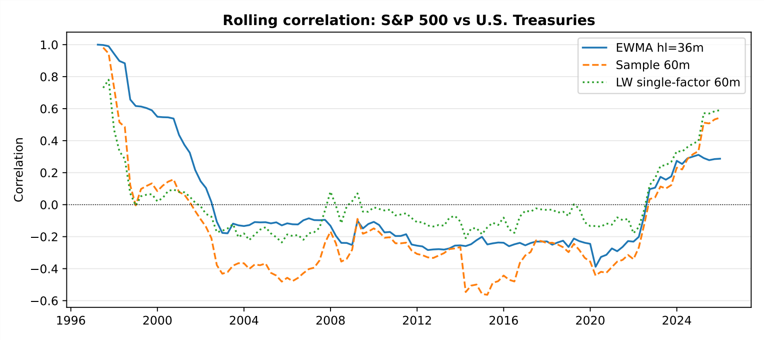


Key insight

EWMA predicts variance best — it adapts to the regime. Shrinkage and the factor model instead pay off on the *correlation* structure.

Rolling correlations

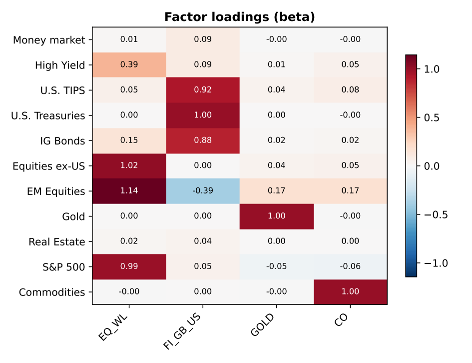
The equity/Treasury correlation is not constant. Single-factor shrinkage gives a smoother, more stable path than a raw 60-month sample window.



Source: book/slides/notebooks/AssetAllocation3_Covariance.ipynb

Factor loadings

A four-factor model compresses the covariance into an $N \times K$ loading matrix.



Source: book/slides/notebooks/AssetAllocation3_Covariance.ipynb

What can go wrong

Garbage in, leverage out

A noisy $\widehat{\Sigma}$ feeds a noisy $\widehat{\Sigma}^{-1}$, which the optimiser turns into large offsetting positions — the covariance is as dangerous an input as expected returns.

- Forgetting to unsmooth illiquid assets understates their risk and over-allocates to them
- A shrinkage target mismatched to the universe ($\alpha^\star \to 0$) buys nothing

Key takeaways

- The sample covariance matrix is noisy and near-singular when $N \sim T$
- Factor structure and shrinkage trade a little bias for a large variance reduction
- Match the shrinkage target to the universe; constant correlation suits one asset class, not many
- Unsmooth illiquid-asset returns before estimating risk
- A better-conditioned Σ yields more stable, less concentrated portfolios

Further reading

Fabozzi et al. (2007) Chapter 8: shrinkage.

Ledoit and Wolf (2004) The well-conditioned shrinkage estimator.

Ledoit and Wolf (2022) A modern review and guide to (non-)linear shrinkage.

Engle et al. (2019) Dynamic conditional correlation for large covariance matrices.

Geltner (1991) Unsmoothing appraisal-based returns.

Backtesting Multi-Asset Strategies

Learning objectives

After this lecture you will be able to:

- Set up a walk-forward backtest of a systematic multi-asset strategy
- Distinguish geometric, arithmetic, and continuously compounded returns and know when each applies
- Evaluate a strategy with performance and trade statistics, not just average return
- Recognise how multiple testing inflates apparent performance

Outline

1. Strategy backtesting and the walk-forward method
2. Cross-validation and resampling in backtesting
3. Geometric vs. arithmetic vs. continuous returns
4. Portfolio performance and trade evaluation
5. Basics of portfolio optimisation; computing covariance
6. Worked example: a minimum-variance portfolio

Computing returns from data

- A simple (discrete) return is the percentage change in prices:

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}}$$

  Simple returns exhibit **portfolio additivity** $R_p = \boldsymbol{w}^\top R$.

- A continuously compounded (logarithmic) return is the log of gross return:

$$r_t = \log(1 + R_t) = \log \frac{P_t}{P_{t-1}} = \log P_t - \log P_{t-1}.$$

  Continuous returns exhibit **time additivity**

  $r_{t:t+2} = r_{t+1} + r_{t+2} = \log P_{t+2} - \log P_t$.

Computing averages

- An arithmetic mean is the simple average:

$$\mu_{\text{arith}} = \frac{1}{T}\sum_{t=1}^{T} R_t$$

  Average returns weight linearly to a portfolio return.

- A geometric mean matches the realised compounded growth:

$$\mu_{\text{geom}} = \left[\prod_{t=1}^{T}(1+R_t)\right]^{1/T} - 1 = \exp\!\left(\tfrac{1}{T}\sum_{t=1}^{T} r_t\right) - 1$$

Key insight

Arithmetic minus geometric $\approx \frac{1}{2}\sigma^2$ — the **variance drag**.

From frictionless aggregation to a real backtest

The textbook portfolio return $R_{p,t} = \boldsymbol{w}^\top R_t$ silently assumes *costless, instantaneous* rebalancing at every observation.

Two frictions break this assumption:

1. Weight drift between rebalance dates as asset prices move.
2. Transaction costs when trading the gap back to target.

A backtest engine must drift the holdings forward, then on each rebalance date ask the strategy for new target weights and charge the trade cost.

Code used in this lecture

The companion notebook `AssetAllocation1_BacktestReturns.ipynb` uses two modules of the repository:

`backtesting/`
: `Backtester(config).run(prices, strategy)` runs the loop; `BacktestConfig` sets initial capital, one-way TC in bps, and rebalance frequency; `BacktestResult` returns the `nav`, `weights_history`, and `turnover`.

`analytics/`
: `TimeseriesAnalyzer(prices, risk_free_col)` derives returns and exposes CAGR, volatility, Sharpe, Sortino, max drawdown, VaR, and CVaR via `apply_standard_functions()`.

A `Strategy` is anything with the signature:

Constant-weight backtest — worked example

S&P 500 / U.S. Treasuries mixes, annually rebalanced, 10 bps one-way TC, 2005–2025 monthly data.

Portfolio	CAGR	Vol	Sharpe	MaxDD
0/100 EQ/FI	2.8%	4.5%	0.12	-18.3%
20/80 EQ/FI	4.7%	4.6%	0.53	-15.4%
40/60 EQ/FI	6.5%	6.4%	0.66	-17.8%
60/40 EQ/FI	8.1%	9.0%	0.66	-27.9%
80/20 EQ/FI	9.6%	12.0%	0.64	-39.7%
100/0 EQ	10.9%	15.1%	0.61	-50.6%

Adding equity raises CAGR throughout, but risk-adjusted return peaks at a balanced 40/60–60/40 mix — the diversification benefit — before drawdowns

What can go wrong

Backtest overfitting

With enough trials, some strategy will look excellent in sample by chance.
Walk-forward discipline and a haircut for the number of trials are not optional.

Harvey et al. (2016) document how the multiple-testing problem pervades the published cross-section of returns; Menkveld et al. (2024) show that even careful researchers reach materially different results on identical data.

Key takeaways

- A credible backtest is walk-forward and out-of-sample by construction
- Return conventions and rebalancing assumptions change the headline numbers
- Treat backtested performance as an upper bound, discounted for the number of trials

Further reading

López de Prado (2018) Chapters 10–15: cross-validation, backtesting, and the dangers of overfitting.

Pardo (2011) Chapters 3–6: walk-forward analysis in detail.

Harvey et al. (2016) Multiple testing and the credibility of factors.

Cirulli et al. (2026) A practical guide to portfolio construction and evaluation.

Coding Session 1

Session objectives

By the end of this hands-on session you will have coded:

- Discrete vs. continuous returns; geometric vs. arithmetic means
- Variance and covariance, by formula and by matrix algebra; annualisation
- Covariance *estimators* — sample, shrinkage, EWMA, and a factor model — compared on their ability to predict future variance
- Portfolio return and portfolio variance from weights
- A long-term constant-weight backtest with periodic rebalancing
- Risk diagnostics: VaR, CVaR, skewness, kurtosis, maximum drawdown

Code in the repo

We build on the framework's backtest engine, analytics, and risk models:

- `backtesting/` `Backtester`, `BacktestConfig`, `BacktestResult`
- `analytics/` `TimeseriesAnalyzer` — CAGR, volatility, Sharpe, Sortino, max drawdown
- `risk_models/` `compute_covariance`, `compute_rolling_covariance`, `compute_rolling_asset_factor_cov`

A minimum-variance portfolio is the running example, carried into Lecture 7. The covariance walk-through (sample vs. shrinkage vs. EWMA vs. factor model) feeds the results in Lecture 3. *Source: book/slides/notebooks/AssetAllocation3_Covariance.ipynb*

Long-Term Expected Returns

Learning objectives

After this lecture you will be able to:

- Explain why long-run forecasts beat trailing average returns as optimiser inputs
- Derive equilibrium (reverse-optimised) expected returns
- Build expected returns block-by-block for fixed income, FX, and equities
- Reason about expected returns for non-yielding assets (gold, crypto)

Outline

1. The equilibrium approach
2. Building-block approach for fixed income and foreign exchange
3. Equity expected returns and the building-block (GKS) model
4. Non-yielding assets: gold and crypto
5. Yield vs. risk-premium framing

Example: Chapter 2 of)

Constructing equilibrium expected returns:

- Estimate market risk premium, risk-free rate, betas
- Expected return = risk-free + market premium × beta

Example US Equities

Expected return of risk-free asset is 3.5%, expected return of the market is 7.5%, and the beta of US equities is 1.33. The equilibrium expected return is $3.5\% + 1.33 \times (7.5\% - 3.5\%) = 8.8\%$.

Key takeaways

- Historical averages are noisy; structure (equilibrium, building blocks) raises forecast R^2
- Fixed-income expected returns decompose into real rate, inflation, term, and credit
- Equity returns build from yield, growth, and valuation change (the GKS supply model)

Further reading

Ilmanen (2012) Expected returns across the major asset classes.

Grinold et al. (2011) The supply-side (GKS) model of the equity premium.

Ferreira and Santa-Clara (2011) The sum-of-the-parts approach to forecasting equity returns.

Kinlaw et al. (2021) Chapter 13: capital-market expectations in practice.

Robust Optimization

Learning objectives

After this lecture you will be able to:

- Compare volatility, VaR, and CVaR as risk measures and objectives
- Cast CVaR optimisation as a linear program
- Explain why MVO is unstable and how resampling regularises it
- Construct a risk-budgeting / risk-parity portfolio

Outline

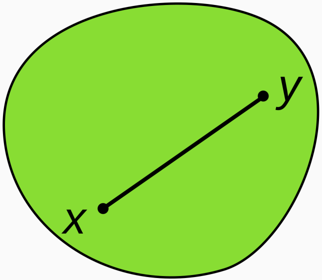
1. Risk measures: volatility, VaR, CVaR
2. CVaR optimisation as a linear program
3. Instability of mean-variance optimisation
4. Resampled robust optimisation
5. Diversification and risk budgeting
6. Risk parity (equal risk contribution)

Convex optimization

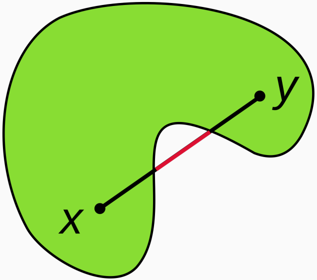
- Minimising a convex function over a convex set.
- Any local optimum is automatically the global optimum, so the problem solves reliably and at scale.
- The workhorse of this lecture: mean–variance, CVaR, and risk-budgeting problems are all convex.

Convex sets and functions

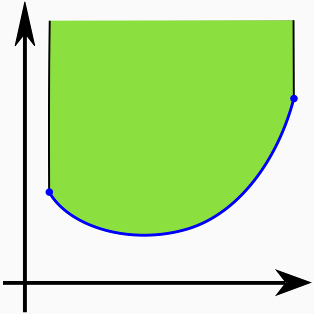
- Convex set: for any $x, y \in C$ and $\theta \in [0,1]$, $\theta x + (1-\theta)y \in C$ — the segment between two points stays inside the set.
- Convex function: $f(\theta x + (1-\theta)y) \leq \theta f(x) + (1-\theta)f(y)$ — equivalently, its epigraph is a convex set.



Convex set



Non-convex set



Convex function

Some risk measures

Write the portfolio loss as $L(\boldsymbol{w}) = -\boldsymbol{w}^\top R$. α is confidence level. Common risk measures:

- Volatility: $\sigma(\boldsymbol{w}) = \sqrt{\boldsymbol{w}^\top \Sigma \boldsymbol{w}}$
- Value-at-risk: $\mathrm{VaR}_\alpha(\boldsymbol{w}) = \inf\{\ell : \Pr\{L(\boldsymbol{w}) \leq \ell\} \geq \alpha\}$
- Expected shortfall (CVaR): $\mathrm{ES}_\alpha(\boldsymbol{w}) = \mathbb{E}\left[L(\boldsymbol{w}) \mid L(\boldsymbol{w}) \geq \mathrm{VaR}_\alpha(\boldsymbol{w})\right]$

Key insight

VaR answers "how bad on a normal bad day"; CVaR answers "how bad on the bad days beyond that." CVaR is coherent and convex — VaR is neither.

Gaussian risk measures

Assume asset returns are normally distributed: $R \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. Φ and ϕ are normal CDF and PDF, respectively. The risk measures then admit closed-form expressions:

$$\begin{aligned}
\sigma(\boldsymbol{w}) &= \sqrt{\boldsymbol{w}^\top \boldsymbol{\Sigma}\, \boldsymbol{w}} \\
\mathrm{SD}_c(\boldsymbol{w}) &= -\boldsymbol{w}^\top \boldsymbol{\mu} + c\sqrt{\boldsymbol{w}^\top \boldsymbol{\Sigma}\, \boldsymbol{w}} \\
\mathrm{VaR}_\alpha(\boldsymbol{w}) &= -\boldsymbol{w}^\top \boldsymbol{\mu} + \Phi^{-1}(\alpha)\sqrt{\boldsymbol{w}^\top \boldsymbol{\Sigma}\, \boldsymbol{w}} \\
\mathrm{ES}_\alpha(\boldsymbol{w}) &= -\boldsymbol{w}^\top \boldsymbol{\mu} + \frac{\phi\big(\Phi^{-1}(\alpha)\big)}{1-\alpha}\sqrt{\boldsymbol{w}^\top \boldsymbol{\Sigma}\, \boldsymbol{w}}
\end{aligned}$$

All four are affine in $\boldsymbol{\mu}$ and linear in $\sqrt{\boldsymbol{w}^\top \boldsymbol{\Sigma}\, \boldsymbol{w}}$ — minimising any of them yields the same efficient frontier, only the risk-aversion coefficient changes.

Gaussian risk contributions

- Volatility $\sigma(\boldsymbol{w})$:
$$\mathrm{RC}_i = w_i \, \frac{(\Sigma \boldsymbol{w})_i}{\sqrt{\boldsymbol{w}^\top \Sigma \, \boldsymbol{w}}}$$
- Standard-deviation measure $\mathrm{SD}_c(\boldsymbol{w})$:
$$\mathrm{RC}_i = w_i \left(-\mu_i + c \, \frac{(\Sigma \boldsymbol{w})_i}{\sqrt{\boldsymbol{w}^\top \Sigma \, \boldsymbol{w}}} \right)$$
- Value-at-risk $\mathrm{VaR}_\alpha(\boldsymbol{w})$:
$$\mathrm{RC}_i = w_i \left(-\mu_i + \Phi^{-1}(\alpha) \, \frac{(\Sigma \boldsymbol{w})_i}{\sqrt{\boldsymbol{w}^\top \Sigma \, \boldsymbol{w}}} \right)$$
- Expected shortfall $\mathrm{ES}_\alpha(\boldsymbol{w})$:
$$\mathrm{RC}_i = w_i \left(-\mu_i + \frac{\phi(\Phi^{-1}(\alpha))}{1 - \alpha} \, \frac{(\Sigma \boldsymbol{w})_i}{\sqrt{\boldsymbol{w}^\top \Sigma \, \boldsymbol{w}}} \right)$$

CVaR optimisation is a linear program

Rockafellar and Uryasev (2002) show CVaR can be minimised by the convex auxiliary function

$$F_\alpha(\boldsymbol{w}, \beta) = \beta + \frac{1}{1-\alpha}\,\mathbb{E}\big[(-\boldsymbol{w}^\top R - \beta)^+\big]$$

With T scenarios the expectation becomes a sum, giving a linear program:

$$\min_{\boldsymbol{w},\beta}\ \beta + \frac{1}{(1-\alpha)T}\sum_{t=1}^{T} u_t \quad \text{s.t.} \quad u_t \geq 0, \quad u_t \geq -\boldsymbol{w}^\top R_t - \beta$$

Solvable at scale with off-the-shelf LP solvers.

Mean-variance optimisation is unstable

Three assets, $\mu = (8, 8, 5)\%$, $\sigma = (20, 21, 10)\%$, $\rho = 80\%$, target volatility 15%. The optimum swings wildly under tiny input changes:

Weight (%)	Base	σ_2=18	μ_1=9
x_1	38.3	13.7	60.6
x_2	20.2	56.1	-5.4
x_3	41.5	30.2	44.8

A 1% change in μ_1 flips x_2 from $+20\%$ to -5%; cutting σ_2 by 3 points nearly triples x_2.

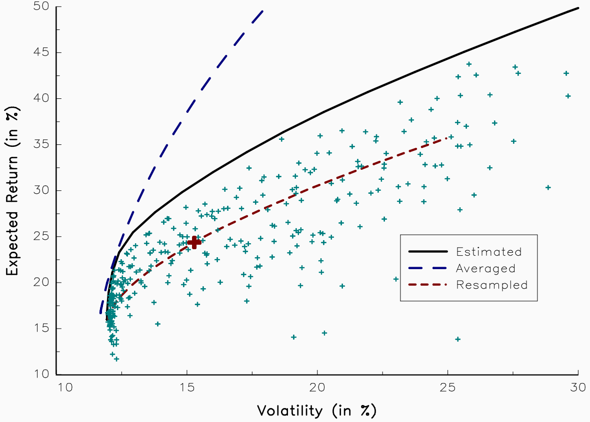
Regularisation: the menu

To stabilise the optimal portfolio we regularise the inputs or the weights:

- Resampling techniques (jackknife, cross-validation, bootstrap)
- Factor analysis and shrinkage (Lecture 3)
- Random matrix theory
- Norm penalisation on the weights

All push the solution away from the corners that pure MVO loves.

Resampled optimisation



Michaud and Michaud (2007): resample inputs, optimise each draw, then average the weights.

The averaged frontier is smoother and the weights far less extreme — an ensemble over plausible inputs rather than a bet on one point estimate.

Source: Course slide figure pool

Resampling techniques: a recipe

1. Draw T samples from $\mathcal{N}(\hat{\boldsymbol{\mu}}, \hat{\Sigma})$ and re-estimate $\hat{\boldsymbol{\mu}}_i,\ \hat{\Sigma}_i$.
2. Optimise on $\hat{\boldsymbol{\mu}}_i, \hat{\Sigma}_i$: find σ_{GMV} and σ_{MR}, partition $[\sigma_{\mathrm{GMV}}, \sigma_{\mathrm{MR}}]$ into M points, and solve for the max-return portfolios $\boldsymbol{w}_{1,i}, \dots, \boldsymbol{w}_{M,i}$.
3. Repeat steps 1–2 a total of I times.

The resampled portfolio at risk level m is the cross-draw average

$$\overline{\boldsymbol{w}}_m = \frac{1}{I} \sum_{i=1}^{I} \boldsymbol{w}_{m,i}$$

Robust portfolio optimisation: formulation

The mean–variance investor solves

$$\max_{\boldsymbol{w}} \ \boldsymbol{w}^\top \boldsymbol{\mu} - \lambda \, \boldsymbol{w}^\top \boldsymbol{\Sigma} \, \boldsymbol{w} \quad \text{s.t.} \quad \boldsymbol{w}^\top \mathbf{1} = 1$$

Treat $\boldsymbol{\mu}$ as uncertain inside an ellipsoidal set around the point estimate $\hat{\boldsymbol{\mu}}$:

$$\mathcal{U}_\delta(\hat{\boldsymbol{\mu}}) = \left\{ \boldsymbol{\mu} \;\middle|\; (\boldsymbol{\mu} - \hat{\boldsymbol{\mu}})^\top \boldsymbol{\Sigma}_\mu^{-1} (\boldsymbol{\mu} - \hat{\boldsymbol{\mu}}) \le \delta^2 \right\}$$

The robust max–min mean–variance problem is then

$$\max_{\boldsymbol{w}} \ \min_{\boldsymbol{\mu} \in \mathcal{U}_\delta(\hat{\boldsymbol{\mu}})} \ \boldsymbol{w}^\top \boldsymbol{\mu} - \lambda \, \boldsymbol{w}^\top \boldsymbol{\Sigma} \, \boldsymbol{w} \quad \text{s.t.} \quad \boldsymbol{w}^\top \mathbf{1} = 1$$

The covariance can also be uncertain, e.g. $\mathcal{U} = \left\{ \boldsymbol{\Sigma} \;\middle|\; \|\boldsymbol{\Sigma} - \tilde{\boldsymbol{\Sigma}}\| \le \Delta\boldsymbol{\Sigma} \right\}$ around a nominal $\tilde{\boldsymbol{\Sigma}}$ in some matrix norm (e.g. Frobenius).

Robust portfolio optimisation: intuition

Solve the inner worst-case problem first:

$$\min_{\boldsymbol{\mu} \in \mathcal{U}_\delta(\hat{\boldsymbol{\mu}})} \; \boldsymbol{w}^\top \boldsymbol{\mu} - \lambda\, \boldsymbol{w}^\top \boldsymbol{\Sigma}\, \boldsymbol{w}$$

The Lagrangian KKT conditions give the worst-case $\boldsymbol{\mu}$ and multiplier

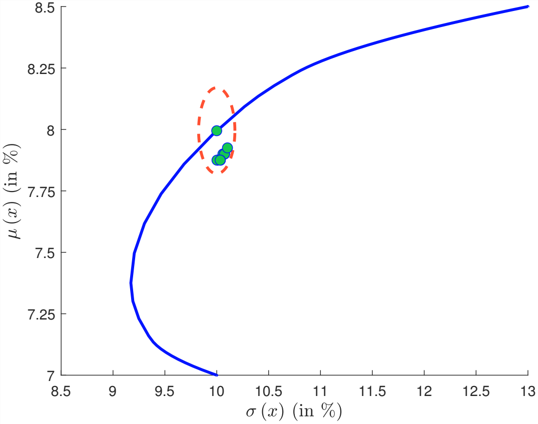
$$\boldsymbol{\mu}^* = \hat{\boldsymbol{\mu}} - \tfrac{1}{2\gamma}\, \boldsymbol{\Sigma}_\mu \boldsymbol{w}, \qquad \gamma^* = \frac{1}{2\delta}\sqrt{\boldsymbol{w}^\top \boldsymbol{\Sigma}_\mu\, \boldsymbol{w}}$$

Substituting $\boldsymbol{\mu}^*, \gamma^*$ back collapses the max–min to one maximisation

$$\max_{\boldsymbol{w}} \; \boldsymbol{w}^\top \hat{\boldsymbol{\mu}} - \lambda\, \boldsymbol{w}^\top \boldsymbol{\Sigma}\, \boldsymbol{w} - \delta\, \sqrt{\boldsymbol{w}^\top \boldsymbol{\Sigma}_\mu\, \boldsymbol{w}}$$

The penalty $\delta\sqrt{\boldsymbol{w}^\top \boldsymbol{\Sigma}_\mu\, \boldsymbol{w}}$ prices estimation risk; δ is the investor's aversion to it.

Optimised vs. diversified



Key insight

Many portfolios with *very different* weights have *nearly identical* mean-variance characteristics. The balanced ones are more robust than the corner MVO solution.

Source: Course slide figure pool

Weight budgeting vs. risk budgeting

For a coherent, convex risk measure, Euler's theorem decomposes total risk into additive contributions:

$$\mathcal{R}(\boldsymbol{w}) = \sum_{i=1}^{N} \underbrace{w_i \frac{\partial \mathcal{R}(\boldsymbol{w})}{\partial w_i}}_{\mathrm{RC}_i}$$

Given budgets $b_i \geq 0$, $\sum_i b_i = 1$:

- Weight budgeting: $w_i = b_i$
- Risk budgeting: $\mathrm{RC}_i = b_i \, \mathcal{R}(\boldsymbol{w})$

Computing the marginal volatilities

The vector of marginal volatilities follows from the chain rule:

$$\begin{aligned}
\frac{\partial\,\sigma(\boldsymbol{w})}{\partial\,\boldsymbol{w}} &= \frac{\partial}{\partial\,\boldsymbol{w}}\left(\boldsymbol{w}^\top \Sigma\,\boldsymbol{w}\right)^{1/2} \\
&= \tfrac{1}{2}\left(\boldsymbol{w}^\top \Sigma\,\boldsymbol{w}\right)^{-1/2}\left(2\,\Sigma\boldsymbol{w}\right) \\
&= \frac{\Sigma\boldsymbol{w}}{\sqrt{\boldsymbol{w}^\top \Sigma\,\boldsymbol{w}}}
\end{aligned}$$

The marginal volatility of asset i is therefore

$$\frac{\partial\,\sigma(\boldsymbol{w})}{\partial w_i} = \frac{(\Sigma\boldsymbol{w})_i}{\sqrt{\boldsymbol{w}^\top \Sigma\,\boldsymbol{w}}} = \sum_{j=1}^{N} \frac{\rho_{ij}\,\sigma_i\sigma_j\,w_j}{\sigma(\boldsymbol{w})} = \sigma_i \sum_{j=1}^{N} w_j\,\frac{\rho_{ij}\sigma_j}{\sigma(\boldsymbol{w})}$$

Computing the risk contributions

Multiplying the marginal volatility by the position size gives the risk contribution of asset i:

$$\begin{aligned}\mathrm{RC}_i &= w_i \cdot \frac{\partial\, \sigma(\boldsymbol{w})}{\partial w_i} \\ &= \frac{w_i\, (\Sigma \boldsymbol{w})_i}{\sqrt{\boldsymbol{w}^\top \Sigma\, \boldsymbol{w}}} \\ &= \sigma_i\, w_i \sum_{j=1}^{N} w_j\, \frac{\rho_{ij}\sigma_j}{\sigma(\boldsymbol{w})}\end{aligned}$$

The Euler allocation principle

Volatility satisfies the full-allocation property:

$$\sum_{i=1}^{N} \mathrm{RC}_i = \sum_{i=1}^{N} \sigma_i w_i \sum_{j=1}^{N} w_j \frac{\rho_{ij}\sigma_j}{\sigma(\boldsymbol{w})} = \frac{1}{\sigma(\boldsymbol{w})} \sum_{i=1}^{N} \sum_{j=1}^{N} w_i w_j \, \rho_{ij}\sigma_i\sigma_j$$

$$= \frac{\sigma^2(\boldsymbol{w})}{\sigma(\boldsymbol{w})} = \sigma(\boldsymbol{w})$$

A shorter proof uses the inner-product form $a \cdot b = a^\top b$:

$$\sum_{i=1}^{N} \mathrm{RC}_i = \frac{\sum_{i=1}^{N} w_i \, (\Sigma \boldsymbol{w})_i}{\sqrt{\boldsymbol{w}^\top \Sigma \, \boldsymbol{w}}} = \frac{\boldsymbol{w}^\top \Sigma \, \boldsymbol{w}}{\sqrt{\boldsymbol{w}^\top \Sigma \, \boldsymbol{w}}} = \sigma(\boldsymbol{w})$$

Risk contribution: definitions

Definitions

Marginal risk contribution of asset i:

$$\mathrm{MR}_i = \frac{\partial\,\sigma(\boldsymbol{w})}{\partial w_i} = \frac{(\Sigma\boldsymbol{w})_i}{\sqrt{\boldsymbol{w}^\top\Sigma\,\boldsymbol{w}}}$$

Absolute risk contribution of asset i:

$$\mathrm{RC}_i = w_i\,\frac{\partial\,\sigma(\boldsymbol{w})}{\partial w_i} = \frac{w_i\,(\Sigma\boldsymbol{w})_i}{\sqrt{\boldsymbol{w}^\top\Sigma\,\boldsymbol{w}}}$$

Relative risk contribution of asset i:

$$\mathrm{RC}_i^\star = \frac{\mathrm{RC}_i}{\sigma(\boldsymbol{w})} = \frac{w_i\,(\Sigma\boldsymbol{w})_i}{\boldsymbol{w}^\top\Sigma\,\boldsymbol{w}}$$

Definition of RB portfolios

Definition

A risk-budgeting (RB) portfolio $\boldsymbol{w}$ satisfies

$$\begin{cases} \mathrm{RC}_1 = b_1\,\mathcal{R}(\boldsymbol{w}) \\ \quad\vdots \\ \mathrm{RC}_i = b_i\,\mathcal{R}(\boldsymbol{w}) \\ \quad\vdots \\ \mathrm{RC}_N = b_N\,\mathcal{R}(\boldsymbol{w}) \end{cases}$$

where $\mathcal{R}(\boldsymbol{w})$ is a coherent, convex risk measure and $b = (b_1, \ldots, b_N)^\top$ is a vector of risk budgets with $b_i \geq 0$ and $\sum_{i=1}^N b_i = 1$.

Risk parity

For the volatility measure the risk contribution of asset i is

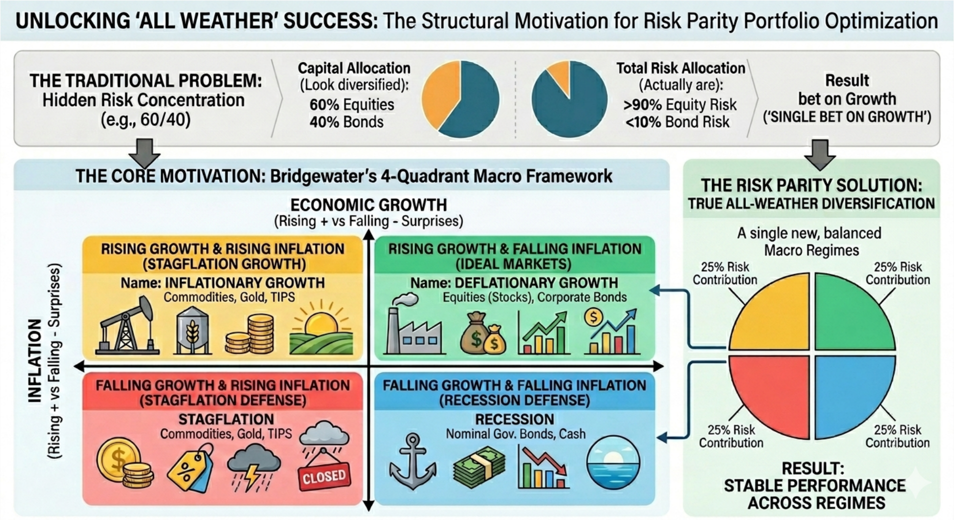
$$\mathrm{RC}_i = w_i \frac{(\boldsymbol{\Sigma w})_i}{\sqrt{\boldsymbol{w}^\top \boldsymbol{\Sigma w}}}$$

Risk parity (equal risk contribution) sets every budget equal, $b_i = 1/N$:

$$\mathrm{RC}_1 = \mathrm{RC}_2 = \cdots = \mathrm{RC}_N$$

No expected returns required — the portfolio depends on $\boldsymbol{\Sigma}$ alone.

Allweather: a risk-parity portfolio



What can go wrong

Robustness is not free

Resampling and risk parity buy stability by *ignoring* information — risk parity discards expected returns entirely, which is a strong assumption.

- Risk parity can lever low-volatility assets heavily to equalise contributions
- CVaR optimisation is only as good as the scenario set $\{R_t\}$ feeding it

Key takeaways

- CVaR is a coherent, convex tail-risk measure and optimises as a linear program
- Plain MVO is an error maximiser: tiny input changes flip the weights
- Resampling averages over plausible inputs, smoothing the frontier and the weights
- Risk budgeting targets risk *contributions*; risk parity equalises them and needs only Σ

Further reading

Rockafellar and Uryasev (2002) CVaR for general loss distributions.

Michaud and Michaud (2007) The resampling solution to estimation error.

Scherer (2007) A critical look at whether robust optimisation helps.

Kuhn et al. (2019) Distributionally robust optimisation.

Views on Markets

Learning objectives

After this lecture you will be able to:

- Recover equilibrium expected returns by reverse-optimising a reference portfolio
- Encode absolute and relative views as a pick matrix with confidence levels
- State the Black–Litterman posterior expected returns
- Explain why Black–Litterman portfolios are better behaved than raw MVO

Outline

1. Motivation: blending equilibrium with views
2. Implied (equilibrium) risk premia
3. Specifying views: the pick matrix
4. The Black–Litterman posterior
5. Optimising with the blended returns

Two steps, two authors

The problem

How do we fold a portfolio manager's tactical views into a strategic allocation without the wild swings of plain mean-variance optimisation?

Black–Litterman is a two-step recipe:

1. Reverse a reference allocation into implied risk premia (Sharpe)
2. Re-optimise from premia blended with the manager's views (Markowitz)

Reverse-optimising the Markowitz problem

The Markowitz investor solves

$$\boldsymbol{w}^\star = \arg\min_{\boldsymbol{w}} \tfrac{1}{2}\,\boldsymbol{w}^\top \Sigma\, \boldsymbol{w} - \gamma\, \boldsymbol{w}^\top(\boldsymbol{\mu} - r_f \mathbf{1})$$
$$\text{s.t.}\quad \mathbf{1}^\top \boldsymbol{w} = 1, \qquad \boldsymbol{w} \in \mathcal{C}$$

When the constraints are slack, the first-order condition and its solution are

$$\Sigma\, \boldsymbol{w} - \gamma(\boldsymbol{\mu} - r_f \mathbf{1}) = \mathbf{0} \quad \Longrightarrow \quad \boldsymbol{w}^\star = \gamma\, \Sigma^{-1}(\boldsymbol{\mu} - r_f \mathbf{1})$$

Reverse the question: treat the observed allocation $\boldsymbol{w}_0$ as the optimum and back out the expected returns that justify it,

$$\tilde{\boldsymbol{\mu}} = r_f \mathbf{1} + \tfrac{1}{\gamma}\, \Sigma\, \boldsymbol{w}_0$$

i.e. $\tilde{\boldsymbol{\mu}}$ is the unique vector of expected returns coherent with $\boldsymbol{w}_0$.

Implied risk premium

If the reference portfolio $\boldsymbol{w}_0$ is optimal for some investor, the first-order condition of the Markowitz problem can be run backwards to reveal the expected returns that justify it:

$$\tilde{\pi} = \tilde{\boldsymbol{\mu}} - r_f \mathbf{1} = \frac{1}{\gamma} \Sigma \, \boldsymbol{w}_0$$

Key insight

Instead of forecasting returns, ask: *what returns would make today's market portfolio optimal?* Those equilibrium premia are the prior.

Implied risk aversion

Computing $\tilde{\boldsymbol{\mu}}$ requires the risk-aversion parameter γ (or its inverse $\phi = \gamma^{-1}$).

Starting from the FOC $\boldsymbol{\Sigma}\, \boldsymbol{w}_0 - \gamma(\tilde{\boldsymbol{\mu}} - r_f \mathbf{1}) = \mathbf{0}$ and left-multiplying by $\boldsymbol{w}_0^\top$ (using $\boldsymbol{w}_0^\top \mathbf{1} = 1$):

$$\begin{aligned}
\gamma \left(\tilde{\boldsymbol{\mu}} - r_f \mathbf{1}\right) &= \boldsymbol{\Sigma}\, \boldsymbol{w}_0 \\
\gamma \left(\boldsymbol{w}_0^\top \tilde{\boldsymbol{\mu}} - r_f\right) &= \boldsymbol{w}_0^\top \boldsymbol{\Sigma}\, \boldsymbol{w}_0 \\
\gamma &= \frac{\boldsymbol{w}_0^\top \boldsymbol{\Sigma}\, \boldsymbol{w}_0}{\boldsymbol{w}_0^\top \tilde{\boldsymbol{\mu}} - r_f}
\end{aligned}$$

Inverting,

$$\phi = \frac{\boldsymbol{w}_0^\top \tilde{\boldsymbol{\mu}} - r_f}{\boldsymbol{w}_0^\top \boldsymbol{\Sigma}\, \boldsymbol{w}_0} = \frac{\mathrm{SR}(\boldsymbol{w}_0 \mid r_f)}{\sqrt{\boldsymbol{w}_0^\top \boldsymbol{\Sigma}\, \boldsymbol{w}_0}} = \frac{\mathrm{SR}(\boldsymbol{w}_0 \mid r_f)}{\sigma(\boldsymbol{w}_0)}$$

where $\mathrm{SR}(\boldsymbol{w}_0 \mid r_f)$ is the portfolio's expected Sharpe ratio.

Implied premium in Sharpe form

Substituting the calibrated γ back into the implied-premium formula:

$$\tilde{\boldsymbol{\mu}} = r_f \mathbf{1} + \mathrm{SR}(\boldsymbol{w}_0 \mid r_f) \frac{\Sigma\, \boldsymbol{w}_0}{\sqrt{\boldsymbol{w}_0^\top \Sigma\, \boldsymbol{w}_0}}$$

The implied risk premium $\tilde{\pi} = \tilde{\boldsymbol{\mu}} - r_f \mathbf{1}$ is therefore

$$\tilde{\pi} = \mathrm{SR}(\boldsymbol{w}_0 \mid r_f) \frac{\Sigma\, \boldsymbol{w}_0}{\sqrt{\boldsymbol{w}_0^\top \Sigma\, \boldsymbol{w}_0}}$$

Key insight

The implied premium is the portfolio's Sharpe ratio scaled by the marginal-volatility direction $\Sigma\, \boldsymbol{w}_0 / \sigma(\boldsymbol{w}_0)$. Both factors are observable from $\boldsymbol{w}_0$ and Σ alone — no separate return forecast is required.

Implied risk premium — numerical example

Example

Take a four-asset portfolio with a given covariance matrix Σ and initial allocation $\boldsymbol{w}_0 = (40\%,\ 30\%,\ 20\%,\ 10\%)$.

- The portfolio volatility is $\sigma(\boldsymbol{w}_0) = 15.35\%$.
- The manager targets a portfolio Sharpe ratio of $\mathrm{SR}(\boldsymbol{w}_0 \mid r_f) = 0.25$.
- The Sharpe-form calibration gives $\phi = \mathrm{SR}(\boldsymbol{w}_0 \mid r_f)/\sigma(\boldsymbol{w}_0) = 1.63$.
- With $r_f = 3\%$, the implied expected returns are

$$\tilde{\boldsymbol{\mu}} = \begin{pmatrix} 5.47\% \\ 6.68\% \\ 8.70\% \\ 9.06\% \end{pmatrix}.$$

Specifying views

The manager's views are linear statements about the (uncertain) returns:

$$P\boldsymbol{\mu} = Q + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(\mathbf{0}, \Omega)$$

- P ($k \times N$) picks the assets in each view; k = number of views
- Q ($k \times 1$) is the asserted value of each view
- Ω encodes confidence: small Ω = strong view, large Ω = weak

Absolute and relative views

Absolute view — "asset 1 will return q_1"

$$P = \begin{pmatrix} 1 & 0 & 0 \end{pmatrix}, \qquad Q = q_1$$

Relative view — "asset 1 beats asset 2 by q"

$$P = \begin{pmatrix} 1 & -1 & 0 \end{pmatrix}, \qquad Q = q_{1|2}$$

Stack rows for multiple views; the diagonal of Ω sets each view's uncertainty.

Gaussian conditioning — the tool

For a jointly Gaussian vector

$$\begin{pmatrix} X \\ Y \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \mu_x \\ \mu_y \end{pmatrix}, \begin{pmatrix} \Sigma_{xx} & \Sigma_{xy} \\ \Sigma_{yx} & \Sigma_{yy} \end{pmatrix}\right),$$

the conditional law $Y \mid X = x$ is again Gaussian, $\mathcal{N}(\mu_{y|x}, \Sigma_{yy|x})$, with

$$\mu_{y|x} = \mu_y + \Sigma_{yx}\Sigma_{xx}^{-1}(x - \mu_x), \qquad \Sigma_{yy|x} = \Sigma_{yy} - \Sigma_{yx}\Sigma_{xx}^{-1}\Sigma_{xy}.$$

Black–Litterman is this formula with returns $Y = \boldsymbol{\mu}$ conditioned on views $X = \nu$.

Posterior expected return

Write the views as $\nu = P\boldsymbol{\mu} + \varepsilon$, so $(\boldsymbol{\mu}, \nu)$ is jointly Gaussian with $\mathrm{Cov}(\boldsymbol{\mu}, \nu) = \Gamma P^\top$ and $\mathrm{Var}(\nu) = P\Gamma P^\top + \Omega$. Conditioning on $\nu = Q$:

$$\bar{\boldsymbol{\mu}} = \mathbb{E}[\boldsymbol{\mu} \mid \nu = Q] = \tilde{\boldsymbol{\mu}} + \Gamma P^\top \left(P\Gamma P^\top + \Omega\right)^{-1}(Q - P\tilde{\boldsymbol{\mu}})$$

- First term: the implied (equilibrium) returns $\tilde{\boldsymbol{\mu}}$ — the prior
- Second term: a correction driven by the disequilibrium $Q - P\tilde{\boldsymbol{\mu}}$ between the manager's views and the market

Posterior covariance

The same conditioning gives the posterior (parameter) covariance

$$\bar{\Sigma} = \mathrm{Var}(\boldsymbol{\mu} \mid \nu = Q) = \Gamma - \Gamma P^\top \left(P \Gamma P^\top + \Omega\right)^{-1} P \Gamma$$

Equivalently, by the Woodbury identity,

$$\bar{\Sigma} = \left(\Gamma^{-1} + P^\top \Omega^{-1} P\right)^{-1}$$

Key insight

A precision-weighted blend of the prior Γ and the view covariance Ω: confident views (small Ω) shrink the posterior uncertainty.

Choosing the covariance matrices

Return covariance Σ

In theory the optimiser uses the posterior

$$\Sigma = \bar{\Sigma} = \left(\Gamma^{-1} + P^{\top}\Omega^{-1}P\right)^{-1}$$

In practice one sets $\Sigma = \hat{\Sigma}$, the sample estimate.

Prior scale Γ

A common choice ties it to the return covariance,

$$\Gamma = \tau\,\Sigma$$

The scalar τ can instead be backed out from a target tracking-error volatility.

Optimising with the blended returns

Feed $\bar{\boldsymbol{\mu}}$ into the standard problem:

$$\boldsymbol{w}^\star(\gamma) = \arg\min_{\boldsymbol{w}} \ \tfrac{1}{2}\boldsymbol{w}^\top \boldsymbol{\Sigma} \boldsymbol{w} - \gamma\, \boldsymbol{w}^\top (\bar{\boldsymbol{\mu}} - r_f \mathbf{1}) \quad \text{s.t.} \quad \mathbf{1}^\top \boldsymbol{w} = 1$$

- No view on an asset $\Rightarrow$ its weight stays near the reference
- A confident view tilts the portfolio in that direction, proportionally
- Weights move smoothly with view strength — no corner solutions

Implementation in five steps

1. Estimate $\hat{\Sigma}$ and set $\Sigma = \hat{\Sigma}$
2. From the reference portfolio $\boldsymbol{w}_0$, compute $\phi = \gamma^{-1}$ and the implied returns $\tilde{\boldsymbol{\mu}}$
3. Specify the views via P, Q and Ω
4. Pick $\Gamma = \tau\Sigma$ and compute the posterior mean $\bar{\boldsymbol{\mu}}$
5. Optimise with $\hat{\Sigma}$, $\bar{\boldsymbol{\mu}}$ and γ

Example: specifying the views

We reuse the four-asset portfolio and require positive optimised weights. The manager holds one absolute and one relative view:

$$P = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \end{pmatrix}, \quad Q = \begin{pmatrix} q_1 \\ q_{2\text{-}3} \end{pmatrix}, \quad \Omega = \begin{pmatrix} \varpi_1^2 & 0 \\ 0 & \varpi_{2\text{-}3}^2 \end{pmatrix}$$

with $q_1 = 4\%$, $q_{2\text{-}3} = -1\%$, $\varpi_1 = 10\%$ and $\varpi_{2\text{-}3} = 5\%$.

Example: five scenarios

Starting from the baseline views, we vary one ingredient at a time:

- Case #1: $\tau = 1$ (baseline)
- Case #2: stronger absolute view, $q_1 = 7\%$
- Case #3: weaker views, $\varpi_1 = \varpi_{2\text{-}3} = 20\%$
- Case #4: tighter prior, $\tau = 10\%$
- Case #5: very tight prior, $\tau = 1\%$

Black–Litterman portfolios

	#0	#1	#2	#3	#4	#5
$x_1^\star$	40.0	33.4	51.2	36.4	38.3	39.8
$x_2^\star$	30.0	51.6	39.9	43.0	42.7	32.6
$x_3^\star$	20.0	5.5	0.0	10.9	9.1	17.7
$x_4^\star$	10.0	9.6	8.9	9.8	9.9	10.0
$\sigma(\boldsymbol{w}^\star \mid \boldsymbol{w}_0)$	0.0	3.7	3.7	2.2	2.2	0.5

#0 is the reference; tighter priors (#4, #5) and weaker views (#3) pull the optimum back toward it, shrinking the tracking error.

Calibrating τ to a tracking error

Target a tracking-error volatility $\sigma^\star$ and solve for τ:

- $\sigma^\star = 2\%$ lies between portfolios #4 ($\sigma(\boldsymbol{w}^\star \mid \boldsymbol{w}_0) = 2.18\%$) and #5 ($0.45\%$)
- hence $\tau \in [1\%, 10\%]$; bisection gives $\tau = 5.2\%$

The resulting optimal portfolio is

$$\boldsymbol{w}^\star = (36.80\%,\ 41.83\%,\ 11.58\%,\ 9.79\%)^\top$$

What can go wrong

Confidence is a free parameter

Ω (and the prior scale $\Gamma = \tau\Sigma$) are chosen, not estimated. Set Ω too small and Black–Litterman collapses back to the unstable MVO it was meant to fix.

- Garbage equilibrium $(\boldsymbol{w}_0, \Sigma) \Rightarrow$ garbage prior
- Relative views are usually more defensible than absolute ones

Key takeaways

- Reverse-optimisation turns a reference portfolio into equilibrium premia — the prior
- Views are linear statements $P\boldsymbol{\mu} = Q + \varepsilon$ with confidence Ω
- The posterior $\bar{\boldsymbol{\mu}}$ shrinks the prior toward the views by their relative precision
- Black–Litterman portfolios move smoothly with view strength, curing MVO's corner solutions

Further reading

Fabozzi et al. (2007) Chapter 9: the Black–Litterman framework.

Meucci (2010) Fully flexible views: generalising Black–Litterman beyond Gaussian.

Coding Session 2

Session objectives

By the end of this hands-on session you will have coded, in the GAA framework:

- Risk contributions of a portfolio from $\boldsymbol{w}$ and Σ
- Risk parity (equal risk contribution) and general risk budgeting
- A 1/N vs. risk-parity backtest comparison over time
- Worst-case mean–variance with elliptical uncertainty in $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$
- A mean–variance vs. robust vs. risk-parity bake-off

This is the computational counterpart of the theory in Lecture 7.

Code in the repo

We drive everything through the same optimisation and backtest modules:

`optimization/risk_budget.py` `RiskBudgetOptimizer` — log-barrier risk parity; direct RC-error NLP with costs

`optimization/robust.py` `RobustMeanVarianceOptimizer`, `worst_case_stats`

`optimization/mean_risk.py` `MeanRiskOptimizer` — the classic baseline

`risk_models/risk_analytics.py` `risk_contributions`, risk-contribution figures

Each optimiser plugs into the Coding Session 1 `Backtester` unchanged. *Source: book/slides/notebooks/AssetAllocation4_Robust.ipynb*

From capital weights to risk contributions

Euler's theorem splits portfolio volatility into additive contributions:

$$\sigma(\boldsymbol{w}) = \sqrt{\boldsymbol{w}^\top \Sigma \boldsymbol{w}} = \sum_{i=1}^{N} \underbrace{w_i \frac{(\Sigma \boldsymbol{w})_i}{\sqrt{\boldsymbol{w}^\top \Sigma \boldsymbol{w}}}}_{\mathrm{RC}_i}$$

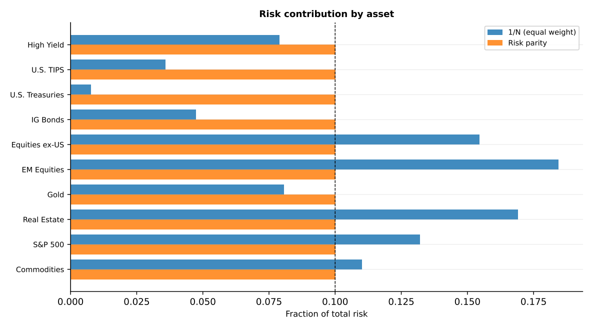
In code, one call returns the RC_i (absolute or relative):

```
risk_contributions(weights, cov, relative=True)
```

Key insight

Equal *capital* weights do not give equal *risk*: the $1/\mathrm{N}$ portfolio quietly loads its risk onto the highest-volatility assets.

1/N vs. risk parity: who carries the risk?

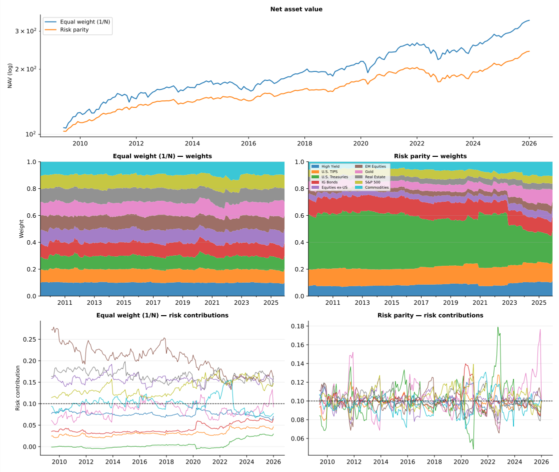


Under $1/\mathrm{N}$ each equity index contributes 15–20% of total risk while Treasuries contribute $\sim$3%.

Risk parity tilts capital toward low-volatility assets so every contribution equals $1/N$ — it needs only Σ, never $\boldsymbol{\mu}$.

Source: AssetAllocation4_Robust.ipynb

Risk parity through time



Annual rebalancing, EWMA covariance (36-month half-life).

Between rebalances the weights drift, so risk-parity contributions fan out from $1/N$ and snap back each year.

Net of costs risk parity cuts volatility ($\sim$5.4% vs. $\sim$8.4%) and drawdowns, lifting the Sharpe ratio ($\sim$0.98 vs. $\sim$0.86) — a steadier ride, even at a lower raw return.

Source: AssetAllocation4_Robust.ipynb

Risk budgeting when trading is costly

Frictionless risk parity solves the scale-free log-barrier program and normalises — giving *exact* budgets:

$$\min_{\boldsymbol{w}} \ \tfrac{1}{2}\,\boldsymbol{w}^\top \Sigma \boldsymbol{w} - \sum_i b_i \ln w_i, \qquad \boldsymbol{w} \leftarrow \boldsymbol{w} / \textstyle\sum_i w_i.$$

But turnover costs act on the *normalised* weights and break that structure. So with costs we optimise the definition of risk budgeting directly:

$$\min_{\boldsymbol{w}} \ \sum_i \left(\mathrm{RC}_i^\star(\boldsymbol{w}) - b_i\right)^2 + \mathrm{tc}\,\|\boldsymbol{w} - \boldsymbol{w}_{\text{cur}}\|_1 \quad \text{s.t.} \quad \boldsymbol{w}^\top \mathbf{1} = 1,\ \boldsymbol{w} \geq 0.$$

Key insight

The RC error is dimensionless, so a basis-point cost trades off on a stable scale — no convexity reformulation, no coefficient tuning. Non-convex, but SLSQP from $1/N$ recovers exact budgets every time.

Worst-case mean–variance: the uncertainty sets

Classic MVO trusts the point estimates $(\hat{\boldsymbol{\mu}}, \hat{\Sigma})$. Robust optimisation hedges over elliptical uncertainty sets around them:

$$\mathcal{U}_\mu = \{\boldsymbol{\mu} : (\boldsymbol{\mu} - \hat{\boldsymbol{\mu}})^\top \Sigma_\mu^{-1} (\boldsymbol{\mu} - \hat{\boldsymbol{\mu}}) \leq k_\mu^2\}, \qquad \mathcal{U}_\Sigma = \{\boldsymbol{\Sigma} : \text{entry-wise ellipsoid}\}$$

`worst_case_stats` sizes them under normality: $\boldsymbol{\Sigma}_\mu = \hat{\Sigma}/T$, with χ^2 radii k_μ, k_σ.

Worst-case return

The inner minimisation over $\mathcal{U}_\mu$ has a closed form:
$\min_{\boldsymbol{\mu} \in \mathcal{U}_\mu} \boldsymbol{w}^\top \boldsymbol{\mu} = \boldsymbol{w}^\top \hat{\boldsymbol{\mu}} - k_\mu \, \|\boldsymbol{\Sigma}_\mu^{1/2} \boldsymbol{w}\|_2$.

Making the covariance worst case tractable

The worst-case *variance* $\max_{\Sigma \in \mathcal{U}_\Sigma} \boldsymbol{w}^\top \Sigma \boldsymbol{w}$ is quadratic in $\boldsymbol{w}$. Lift it with $\mathbf{W} \succeq \boldsymbol{w}\boldsymbol{w}^\top$ (a Schur complement):

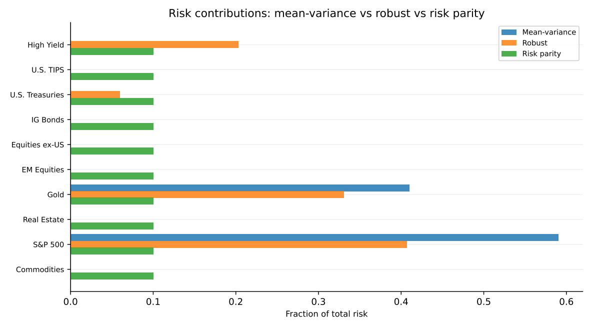
$$\max_{\Sigma \in \mathcal{U}_\Sigma} \boldsymbol{w}^\top \Sigma \boldsymbol{w} \;\le\; \langle \hat{\Sigma}, \mathbf{W} \rangle + k_\sigma \, \| \sqrt{V} \odot \mathbf{W} \|_F, \qquad \begin{bmatrix} \mathbf{W} & \boldsymbol{w} \\ \boldsymbol{w}^\top & 1 \end{bmatrix} \succeq 0$$

This turns the max–min into one small semidefinite program, solved by `cvxpy` (CLARABEL/SCS) — or, in closed form, by `scipy` SLSQP — inside `RobustMeanVarianceOptimizer`.

Scope

We optimise *max-utility* (or min-variance), not max-Sharpe: the robust Sharpe ratio is a fractional objective that does not compose with the SDP lift.

Three philosophies, same inputs



Source: AssetAllocation4_Robust.ipynb

MVO trusts $\hat{\mu}$ — concentrated bets

Robust hedges estimation error

Risk parity ignores μ — spreads risk evenly

Results: robust buys a lower-risk mix

Same 10-asset universe, $\lambda = 5$, full-sample inputs:

	Mean–variance	Robust	Risk parity
Volatility	6.8%	5.8%	4.3%
Return/risk	1.33	1.48	1.41
Eff. # assets	2.8	2.6	6.4

Robust trims expected return but delivers the best worst-case-adjusted return/risk; risk parity is by far the most diversified.

What can go wrong

No free lunch

Robustness buys stability by *discarding* information — risk parity throws away μ entirely, and robust MVO shrinks toward the centre of its uncertainty sets.

- Risk parity levers low-volatility assets hard — exclude near-cash names ($\Sigma_{ii} \to 0$ blows up the weight)
- "More robust" is *not* "more diversified": the robust optimum can still concentrate, just into better-estimated assets
- The uncertainty radii k_μ, k_σ are modelling choices — calibrate them

Key takeaways

- Equal capital weights $\neq$ equal risk; `risk_contributions` makes the gap visible
- Risk parity equalises RC_i from Σ alone — a steadier risk profile through time
- Robust MVO hedges elliptical uncertainty in $\boldsymbol{\mu}$ *and* Σ via a small SDP
- On our data robust earns the best worst-case-adjusted return/risk; risk parity is the most diversified
- One `optimization` API, one `Backtester` — swapping objectives is a one-line change

Further reading

Goldfarb and Iyengar (2003) Robust portfolio selection over ellipsoidal uncertainty sets.

Lobo and Boyd (2000) The worst-case variance lift used here.

Tütüncü and Koenig (2004) Saddle-point robust asset allocation.

Roncalli (2013) The standard reference on risk parity and budgeting.

Spinu (2013) The log-barrier risk-parity solver behind `RiskBudgetOptimizer`.

Macroeconomic and Market Regimes

Learning objectives

After this lecture you will be able to:

- Define a regime as a persistent shift in the joint distribution of returns
- Distinguish macroeconomic regimes from market (risk) regimes
- Detect regimes with turbulence measures and mixture/hidden-Markov models
- Feed a regime into the optimiser through $\boldsymbol{\Sigma}$ and $\boldsymbol{\mu}$

Outline

1. What are regimes and why they matter
2. Macroeconomic regimes
3. Market regimes
4. Detecting regimes
5. Regimes in static optimisation

What is a regime?

A regime is a persistent state in which the joint distribution of returns —means, volatilities, and correlations— is materially different from other states.

Key insight

Correlations are not constant. They rise in crises exactly when diversification is most needed. A single unconditional Σ averages over states that never coexist.

Ang and Bekaert (2002) show that allowing for regime shifts changes the optimal international allocation.

Macroeconomic regimes

Defined from observable macro variables:

Output gap expansion vs. recession — growth above or below potential

Oil energy-price shocks as a supply/inflation driver

Yield curve steep vs. inverted — a classic recession signal

These are interpretable and map to economic narratives, but they are released with a lag and revised.

Market regimes

Defined from market data, often faster-moving than macro:

Turbulence statistically unusual joint returns (see next slide)

Volatility / sentiment risk-on vs. risk-off

Geopolitical event-driven stress

Market regimes lead macro regimes but are noisier — they react to prices, not fundamentals.

Detecting regimes

Financial turbulence flags return vectors far from their historical norm via the Mahalanobis distance:

$$d_t = (r_t - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (r_t - \boldsymbol{\mu})$$

- Large d_t = an unusual combination of returns, not just a large move
- Gaussian mixtures / hidden-Markov models cluster periods into latent regimes and yield a usable regime probability each date
- The framework's `regimes/` module implements these classifiers

Regimes through the covariance matrix

Replace the unconditional covariance with a regime-conditional one:

$$\Sigma \longrightarrow \Sigma_{\text{regime}}$$

- Turbulent-regime Σ has higher volatilities and correlations
- Optimising against it produces a more defensive, less concentrated portfolio
- A regime-weighted blend $\sum_s p_s \Sigma_s$ avoids hard switches

Regimes through expected returns

Condition the return vector on the regime and run scenario analysis:

$$\boldsymbol{\mu} \longrightarrow \boldsymbol{\mu}_{\text{regime}}, \qquad \bar{\boldsymbol{\mu}} = \sum_s p_s \, \boldsymbol{\mu}_s$$

- Tilt the scenario probabilities p_s toward a feared or favoured state
- Re-optimise to see how the allocation responds
- Makes the regime view explicit and auditable, not buried in the inputs

What can go wrong

Regimes are easy to fit, hard to trade

You only know you were in a regime *after* it ends. Detection lags, and each regime has few independent observations to estimate its Σ and $\boldsymbol{\mu}$.

- In-sample regime labels invite look-ahead bias — detect with data available at the time
- Switching too eagerly churns the portfolio and pays transaction costs for noise

Key takeaways

- Regimes capture persistent shifts in the joint return distribution; correlations spike in stress
- Macro regimes are interpretable but lagged; market regimes are fast but noisy
- Turbulence (Mahalanobis distance) and mixture/HMM models detect regimes
- Condition Σ and $\boldsymbol{\mu}$ on the regime; scenario analysis makes the view auditable

Further reading

Ang and Bekaert (2002) International asset allocation with regime shifts.

Kinlaw et al. (2021) Chapters 14 and 22: turbulence and regimes in practice.

Kritzman et al. (2011) Principal components as a measure of systemic risk.

Pedersen (2019) Chapter 11: how market conditions drive returns.

Tactical Asset Allocation and Dynamic Strategies

Learning objectives

After this lecture you will be able to:

- State the goal of TAA and measure short-term outperformance (IR, IC)
- Generate signals from macro, valuation, trend/momentum, and carry
- Build a pro-forma and relative-performance time series for a tactical strategy
- Account for transaction costs when evaluating a dynamic strategy

Outline

1. What is tactical asset allocation?
2. Signal generation and backtesting systematic strategies
3. Macro investing; technical and sentiment analysis
4. Valuations; trend and momentum; value and carry
5. Short-term expected-return prediction

From absolute to active allocation

In Lecture 2 we chose weights $\boldsymbol{w}$ to trade expected return $\boldsymbol{w}^\top \boldsymbol{\mu}$ against absolute risk $\boldsymbol{w}^\top \Sigma \boldsymbol{w}$.

Tactical allocation is judged relative to a benchmark $\boldsymbol{b}$ — the strategic policy mix. We recast the same Markowitz machinery in terms of active return and active risk.

Key insight

Replace expected return with expected *excess* return over $\boldsymbol{b}$, and volatility with *tracking-error* volatility. The optimization is structurally identical — only the inputs are re-centred on the benchmark.

Tracking error

- Portfolio $\boldsymbol{w} = (w_1, \ldots, w_N)^\top$
- Benchmark $\boldsymbol{b} = (b_1, \ldots, b_N)^\top$
- The tracking error between the active portfolio $\boldsymbol{w}$ and its benchmark $\boldsymbol{b}$ is the difference between the return of the portfolio and the return of the benchmark:

$$e = R(\boldsymbol{w}) - R(\boldsymbol{b}) = \sum_{i=1}^{N} w_i R_i - \sum_{i=1}^{N} b_i R_i = \boldsymbol{w}^\top R - \boldsymbol{b}^\top R = (\boldsymbol{w} - \boldsymbol{b})^\top R$$

- The expected excess return is:

$$\mu(\boldsymbol{w} \mid \boldsymbol{b}) = \mathbb{E}[e] = (\boldsymbol{w} - \boldsymbol{b})^\top \boldsymbol{\mu}$$

- The volatility of the tracking error is:

$$\sigma(\boldsymbol{w} \mid \boldsymbol{b}) = \sigma(e) = \sqrt{(\boldsymbol{w} - \boldsymbol{b})^\top \boldsymbol{\Sigma} (\boldsymbol{w} - \boldsymbol{b})}$$

Markowitz optimization problem

The expected return of the portfolio is replaced by the expected excess return, and the volatility of the portfolio by the volatility of the tracking error.

σ-problem

The investor maximizes the expected tracking error subject to a constraint on the tracking-error volatility:

$$\boldsymbol{w}^{\star} = \arg\max \ \mu(\boldsymbol{w} \mid \boldsymbol{b})$$
$$\text{s.t.} \begin{cases} \mathbf{1}^{\top}\boldsymbol{w} = 1 \\ \sigma(\boldsymbol{w} \mid \boldsymbol{b}) \leq \sigma^{\star} \end{cases}$$

Equivalent QP problem

We transform the σ-problem into a γ-problem:

$$\boldsymbol{w}^{\star}(\gamma) = \arg\min f(\boldsymbol{w} \mid \boldsymbol{b})$$

with

$$\begin{aligned} f(\boldsymbol{w} \mid \boldsymbol{b}) &= \tfrac{1}{2}(\boldsymbol{w} - \boldsymbol{b})^{\top} \boldsymbol{\Sigma} (\boldsymbol{w} - \boldsymbol{b}) - \gamma (\boldsymbol{w} - \boldsymbol{b})^{\top} \boldsymbol{\mu} \\ &= \tfrac{1}{2}\boldsymbol{w}^{\top} \boldsymbol{\Sigma} \boldsymbol{w} - \boldsymbol{w}^{\top} (\gamma \boldsymbol{\mu} + \boldsymbol{\Sigma} \boldsymbol{b}) + \left(\tfrac{1}{2}\boldsymbol{b}^{\top} \boldsymbol{\Sigma} \boldsymbol{b} + \gamma \boldsymbol{b}^{\top} \boldsymbol{\mu}\right) \\ &= \tfrac{1}{2}\boldsymbol{w}^{\top} \boldsymbol{\Sigma} \boldsymbol{w} - \boldsymbol{w}^{\top} (\gamma \boldsymbol{\mu} + \boldsymbol{\Sigma} \boldsymbol{b}) + c \end{aligned}$$

where c is a constant which does not depend on portfolio $\boldsymbol{w}$.

QP problem with $Q = \boldsymbol{\Sigma}$ and $R = \gamma \boldsymbol{\mu} + \boldsymbol{\Sigma} \boldsymbol{b}$

Remark

The efficient frontier is the parametric curve $\left(\sigma(\boldsymbol{w}^{\star}(\gamma) \mid \boldsymbol{b}), \mu(\boldsymbol{w}^{\star}(\gamma) \mid \boldsymbol{b})\right)$ with $\gamma \in \mathbb{R}_{+}$

Information ratio

Consider a combination of the benchmark $\boldsymbol{b}$ and the active portfolio $\boldsymbol{w}$:

$$\boldsymbol{y} = (1-\alpha)\boldsymbol{b} + \alpha\boldsymbol{w}, \qquad \alpha \geq 0,$$

where α is the proportion of wealth invested in the portfolio $\boldsymbol{w}$. It follows that

$$\mu(\boldsymbol{y} \mid \boldsymbol{b}) = (\boldsymbol{y} - \boldsymbol{b})^\top \boldsymbol{\mu} = \alpha\,\mu(\boldsymbol{w} \mid \boldsymbol{b})$$

and

$$\sigma^2(\boldsymbol{y} \mid \boldsymbol{b}) = (\boldsymbol{y} - \boldsymbol{b})^\top \boldsymbol{\Sigma}(\boldsymbol{y} - \boldsymbol{b}) = \alpha^2\,\sigma^2(\boldsymbol{w} \mid \boldsymbol{b}).$$

We deduce that

$$\mu(\boldsymbol{y} \mid \boldsymbol{b}) = \mathrm{IR}(\boldsymbol{w} \mid \boldsymbol{b}) \cdot \sigma(\boldsymbol{y} \mid \boldsymbol{b}).$$

The efficient frontier is a straight line

Tangency portfolio

With additional constraints, the portfolio optimization problem becomes:

$$\boldsymbol{w}^\star(\gamma) = \arg\min \; \tfrac{1}{2}\boldsymbol{w}^\top \boldsymbol{\Sigma}\boldsymbol{w} - \boldsymbol{w}^\top(\gamma\boldsymbol{\mu} + \boldsymbol{\Sigma}\boldsymbol{b})$$

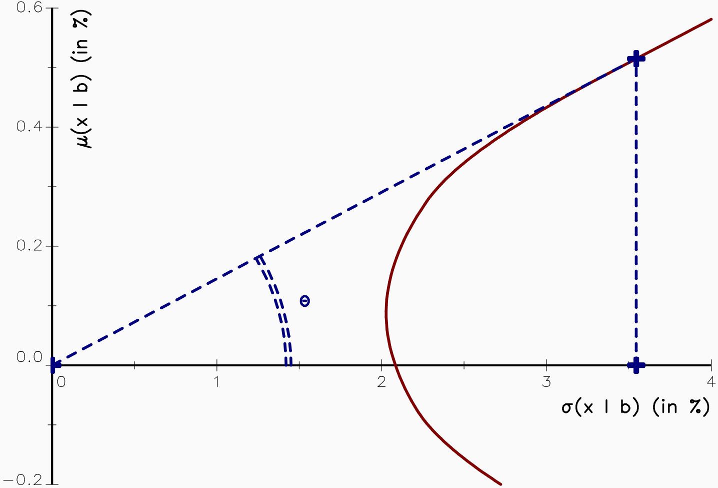
$$\text{s.t.} \quad \begin{cases} \mathbf{1}^\top \boldsymbol{w} = 1 \\ \boldsymbol{w} \in \Omega \end{cases}$$

The efficient frontier is no longer a straight line

Tangency portfolio

One optimized portfolio dominates all the others: it lies on the efficient frontier at the point tangent to the straight line through the benchmark. It is also the portfolio that maximizes the information ratio.

Constrained efficient frontier with a benchmark



The tangency portfolio with respect to a benchmark (Example 3, 3rd case).

What can go wrong

Costs and crowding eat the edge

Short-horizon signals turn over fast. Transaction costs, capacity, and crowding can erase a statistically real signal before it reaches the bottom line.

Key takeaways

- TAA seeks short-horizon outperformance over the strategic policy mix
- Information ratio and information coefficient quantify skill net of risk
- A tactical signal is only useful after transaction costs and capacity constraints

Further reading

Kinlaw et al. (2021) Chapter 17: tactical asset allocation.

Ilmanen et al. (2021) How factor premia vary over time — a century of evidence.

Goodwin (1998) The information ratio.

Bryzgalova et al. (2026) The term structure of macro risk premia.

Backtesting and Tactical Portfolio Construction

Learning objectives

After this lecture you will be able to:

- Build two momentum strategies as explicit portfolio-construction rules
- Turn a single backtest into a distribution of out-of-sample paths
- Test whether the best of many trials is real, not luck, with the probabilistic and deflated Sharpe ratios and the bootstrap reality check
- Read an in-sample number as an upper bound, discounted for the search

Outline

1. Two momentum rules: time-series and cross-sectional
2. One honest path: walk-forward analysis
3. Many paths: combinatorial purged cross-validation
4. Was it luck? PSR, DSR, and the bootstrap reality check / SPA
5. Worked examples: a single asset and three cross-sections

Time-series momentum — a single-asset rule

The position is the sign of the trailing L-period return; it is held over the next period and earns that period's return:

$$w_t = \operatorname{sign}\Big(\textstyle\sum_{i=0}^{L-1} R_{t-i}\Big) \in \{-1, 0, +1\}, \qquad R_{t+1}^{\text{strat}} = w_t\, R_{t+1}.$$

The signal at t uses only data through t, and the weight applies to the *next* return — so the rule is causal by construction. The only free parameter is the lookback L.

Cross-sectional momentum — a long-only rule

Rank the assets by trailing L-period return, then tilt a fully-invested equal-weight book toward the top-quantile winners and away from the bottom-quantile losers (equal weight *within* each leg):

$$w_{i,t} = \frac{1}{n} + \alpha\, o_{i,t}, \qquad o_{i,t} \in \left\{ +\tfrac{1}{k},\ 0,\ -\tfrac{1}{k} \right\}, \qquad \textstyle\sum_i w_{i,t} = 1.$$

Here $k = \lfloor q\, n \rfloor$ and α is the largest scale keeping every weight ≥ 0 (the most-shorted name lands at zero), so the book stays long-only. Two parameters are searched: the lookback L and the quantile q. Long-only makes the tilt directly comparable to an equal-weight benchmark.

Why these rules are a backtesting cautionary tale

Key insight

Each rule is a pure function of one or two parameters. Sweeping the grid and reporting the best is exactly the recipe for an over-fit backtest — which makes momentum the perfect vehicle for studying validation.

The rest of the lecture treats the lookback (and quantile) as something to be *validated*, not merely optimised.

One engine drives every number

A `ParameterGrid` of lookbacks (and quantiles) plus a strategy factory wire each rule into `backtesting.Backtester`. Every cell is backtested once over the full sample and cached:

In-sample pick best-Sharpe cell on the whole sample.

Walk-forward each refit *slices* the cache to its train window.

CPCV each test group reindexes the same causal returns.

Data-snooping the cached cells *are* the trial matrix.

Backtested returns are causal, so one cache serves all four schemes — and costs, weight drift, and rebalancing apply identically everywhere.

Walk-forward analysis

- Expand a training window; pick the lookback that maximises its in-sample Sharpe.
- Apply that choice to the next, disjoint block; record the returns.
- Roll forward and repeat. Concatenating the blocks gives **one** out-of-sample equity path.

Key insight

The backtester *is* the walk-forward loop: a `WalkForwardStrategy` re-selects its lookback inside `compute_weights` on a refit schedule, so a single backtest produces the entire out-of-sample path.

Honest — every return is genuinely out-of-sample — but a *single draw* from a sampling distribution we cannot see.

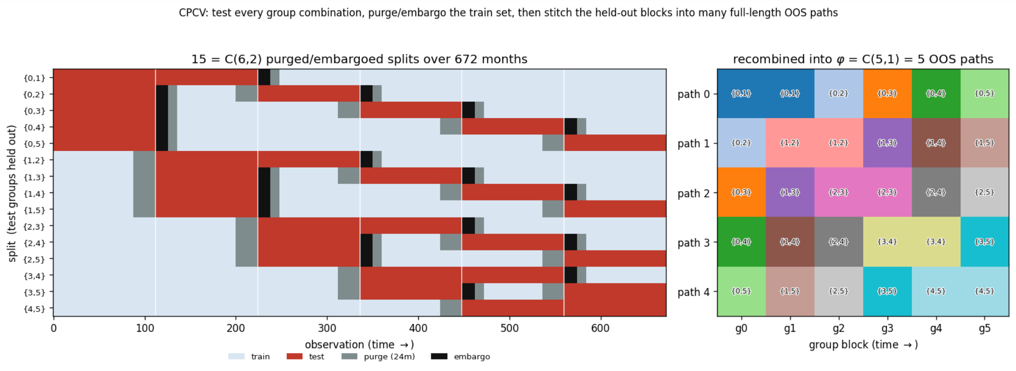
Combinatorial purged cross-validation

Split the timeline into N groups, test every combination of k, and recombine the out-of-sample predictions into many full-length paths:

$$\varphi = \binom{N-1}{k-1} \quad \text{distinct OOS paths.}$$

Each group is tested in φ combinations, so it contributes one segment to each path. The Sharpe of every path is one observation — a distribution, not a point estimate.

The CPCV scheme on US equities



Left: all $\binom{6}{2} = 15$ splits over 672 months — each holds out two groups (red), with the lookback overlap purged (grey) and a short embargo after each block (black); the rest trains. Right: every group is held out in $\varphi = \binom{5}{1} = 5$ splits, so one segment per group stitches into five full-length out-of-sample paths.

Leakage control: purge and embargo

Because test groups can sit *inside* the sample, the training set straddles them in time and would otherwise leak.

Purge drop training points within the signal lookback of either edge of a test block.

Embargo additionally drop a few points right after each test block to absorb residual serial correlation.

Without purging

Adjacent, overlapping windows make the out-of-sample estimate optimistic — the path Sharpes shift up and the dispersion shrinks.

Probabilistic Sharpe ratio

The probability that the *true* Sharpe beats a benchmark SR^*, given the estimate $\widehat{SR}$ from T returns with skew γ_3 and kurtosis γ_4:

$$\mathrm{PSR}(SR^*) = \Phi\left(\frac{(\widehat{SR} - SR^*)\sqrt{T-1}}{\sqrt{1 - \gamma_3 \widehat{SR} + \frac{\gamma_4 - 1}{4}\widehat{SR}^2}}\right).$$

The denominator is the Mertens correction: negative skew and fat tails widen the estimator's error and deflate the probability.

Deflated Sharpe ratio

With N trials, the benchmark SR^* is the expected maximum Sharpe attainable by chance under the null — a Gumbel extreme-value term:

$$SR^* = \sqrt{\mathrm{Var}[\widehat{SR}_n]}\,\left[(1-\gamma)\,\Phi^{-1}\!\left(1-\tfrac{1}{N}\right) + \gamma\,\Phi^{-1}\!\left(1-\tfrac{1}{Ne}\right)\right],$$

and $\mathrm{DSR} = \mathrm{PSR}(SR^*)$.

Effective N

Correlated trials (lookbacks 11 and 12 are nearly identical) make the raw grid size overstate the search. Use the *effective* number of independent trials, or the DSR is too harsh.

Reality check and superior predictive ability

Test the best of many rules against a benchmark, with serial dependence preserved by the stationary bootstrap:

$$V = \max_m \sqrt{T}\,\bar{d}_m, \qquad H_0 : \max_m \mathbb{E}[d_m] \leq 0,$$

where d_m is rule m's performance relative to the benchmark. White's reality check reads the p-value off the bootstrap null; Hansen's SPA studentises and re-centres, removing the drag of poor and irrelevant alternatives.

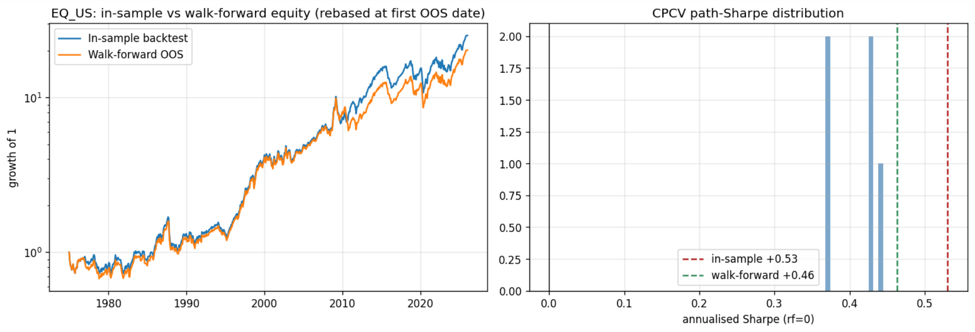
Single asset: time-series momentum on US equities

Lookbacks $\{3, 6, 9, 12, 18, 24\}$ months, monthly data, 10 bps cost.

Stage	Annualised Sharpe / p
In-sample best ($L = 12$)	0.53
Walk-forward (one OOS path)	0.46
CPCV (5 paths, mean $\pm$ sd)	0.41 ± 0.04
Reality check / SPA p	0.001 / 0.001
Deflated Sharpe ratio	0.998

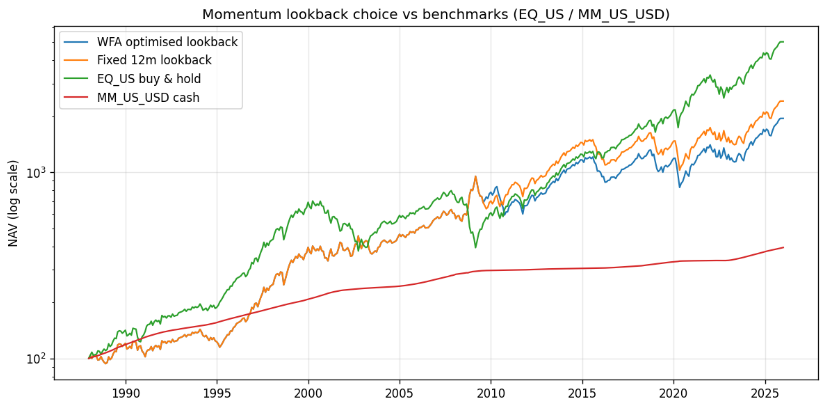
The edge shrinks from in-sample to out-of-sample but survives: trend on US equities is real after the haircut for six trials. *Source: book/slides/notebooks/AssetAllocation5_TAA.ipynb*

Single asset: optimism vs. evidence



Left: the in-sample curve sits above the walk-forward path. Right: the CPCV distribution brackets the honest Sharpe; the in-sample value is its optimistic upper tail. *Source: book/slides/notebooks/AssetAllocation5_TAA.ipynb*

Single asset: trend vs. buy-and-hold



The walk-forward and fixed-12m momentum books trail buy-and-hold EQ_US in raw growth over a secular equity bull market — trend pays for smaller drawdowns, not for beating a rising index. A validated, positive-Sharpe edge is

Cross-section: long-only momentum vs. equal weight

Long-only equal-weight momentum tilt, tested for outperformance over equal weight.

Universe	In-sample SR	OOS SR (WFA)	RC p vs. EW
Macro (10)	1.13	1.10	0.003
US industries (10)	0.87	0.70	0.472
Equity styles (9)	0.97	0.98	0.537

All three have high *absolute* Sharpes — but only the macro cross-section beats equal weight once the search is accounted for. *Source: book/slides/notebooks/AssetAllocation5_TAA.ipynb*

Reading the cross-section result

Key insight

A high Sharpe is not the same as beating the benchmark. Long-only momentum inherits the market premium, so every universe looks good in isolation; only the data-snooping test against equal weight separates skill from beta.

The equity-style result ($p \approx 0.54$) also reflects a short, recent sample — the validation honestly reports low confidence rather than a flattering number.

What can go wrong

The four traps

- **One path**: a single backtest hides its own sampling error — show the CPCV distribution.
- **Multiple testing**: the best of N trials is biased up — deflate it.
- **Effective N**: correlated trials make the naive count too harsh *and* the naive best too generous.
- **Costs and turnover**: short-horizon signals can be real yet unprofitable after trading frictions.

Key takeaways

- Momentum rules are simple, parameterised, and therefore easy to over-fit
- Walk-forward gives one honest path; CPCV gives the distribution around it
- PSR/DSR and the bootstrap reality check turn "looks good" into a p-value
- Treat the in-sample Sharpe as the optimistic upper bound, never the answer

Code used in this lecture

`signals/stcma/momentum.py`
: `TimeSeriesMomentum`, `CrossSectionalMomentum` (engine `Strategys`) and their `*_factory` cell builders.

`signals/ml_pipeline/`
: `signal_research_pipeline` (grid + factory backtest cache), `walk_forward` (`WalkForwardStrategy`), `combinatorial_purged_cv`, `reality_check`.

`analytics/`
: `probabilistic_/deflated_sharpe_ratio_from_returns`; `single_/multi_strategy_report` PDFs.

Drivers: `taa/examples/validation_single_asset.py` and `validation_cross_sectional.py`. *Source: book/slides/notebooks/AssetAllocation5_TAA.ipynb*

Further reading

López de Prado (2018) Combinatorial purged CV and backtest overfitting (ch. 7, 11–12).

Bailey and López de Prado (2014) The deflated Sharpe ratio; Bailey and López de Prado (2012) for the PSR.

White (2000) Reality check; Hansen (2005) the SPA refinement.

Moskowitz et al. (2012) Time-series momentum; Jegadeesh and Titman (1993) cross-sectional momentum.

Coding Session 3

Session objectives

This session is the launchpad for the two group case studies. You will code, in the GAA framework, the building blocks each case needs:

- **Yale** (quarterly): decade-average allocations, a constant-weight backtest, *unsmoothing* of illiquid assets, an ACWI/Agg benchmark, and scenario-tilted expected returns
- **UBS**: four risk *clusters*, a covariance estimate, an equal-risk-contribution portfolio, and volatility targeting via leverage

The notebook solves the plumbing; the *analysis* is your group's work.

Code in the repo

Everything runs locally against the framework — no Colab, no Google Drive:

`backtesting/`	`Backtester` with a constant-weight `Strategy`
`analytics/`	`TimeseriesAnalyzer` — CAGR, vol, Sharpe, drawdown
`data/`	`unsmooth_Geltner` — AR(1) appraisal-filter inversion
`risk_models/`	`compute_covariance` (sample / EWMA)
`optimization/`	`RiskBudgetOptimizer`, `MeanRiskOptimizer`

Public proxies from `gaa_data` stand in for the private Yale series. The Yale case runs at quarterly frequency: illiquid legs (PE, real estate) come from `perf_Q` at their native frequency, the rest are resampled from monthly. *Source: book/slides/notebooks/AssetAllocation6_Cases_GAA.ipynb*

What the Yale case asks

Four questions, increasingly open-ended:

1. Compare the average decade allocations (80s/90s/00s/10s) on performance *and* risk-adjusted returns — after unsmoothing illiquid assets; pick an ACWI/Agg benchmark for each
2. Re-tilt expected returns under the Exhibit-7 COVID scenarios — do you also move Σ? How do allocations shift with the probabilities?
3. How does the 50% illiquidity ceiling constrain manoeuvring room?
4. When does "creating principles, not following a recipe" add value?

Q1–Q2 are coded below; Q3–Q4 are discussion.

Decade allocations as a backtest

Map each Exhibit-1 bucket to a proxy, build a quarterly price index, and run the `Backtester` with constant weights and annual rebalancing over a common window — so the four allocation *styles* meet the same market history.

	1980s	1990s	2000s	2010s
CAGR	9.6%	8.9%	9.0%	9.2%
Volatility	12.2%	11.5%	12.6%	15.3%
Sharpe	0.82	0.81	0.75	0.66

On *reported* returns the lean-alternatives 80s/90s mixes edge the alternatives-heavy 2010s — but the illiquid legs still flatter their risk.

Unsmoothing the illiquid legs

Appraisal / private returns are autocorrelated: reported risk is understated.

Invert the AR(1) appraisal filter (Geltner, 1993) with coefficient ρ:

$$r_t^{\text{true}} = \frac{r_t^{\text{obs}} - \rho\, r_{t-1}^{\text{obs}}}{1-\rho}$$

One call returns the unsmoothed series and ρ:

```
unsmooth_Geltner(returns, max_beta=None)
```

Key insight

At quarterly frequency direct real estate is strongly smoothed ($\rho \approx 0.86$): the real-assets leg *nearly doubles* its volatility once unsmoothed ($\sim$6.9% $\rightarrow$ 11.9%). Risk-adjusted rankings built on *smoothed* data are not to be trusted — recompute the Sharpe after unsmoothing.

A like-for-like benchmark

A defensible yardstick for each decade allocation is a single stock/bond split with the same risk: ACWI $\approx$ `EQ_WL`, Global-Agg $\approx$ `FI_IG_ALL`.

Recipe

Find the equity share x in $x \cdot \text{ACWI} + (1 - x) \cdot \text{Agg}$ whose annualised volatility matches the allocation's, then compare excess return and information ratio.

This separates "did the *mix* help" from "did manager selection help" — the heart of Yale's edge in Exhibit 9.

Scenario-tilted expected returns

Encode each Exhibit-7 scenario as a shift to $\boldsymbol{\mu}$ (and optionally a scaling of Σ), then form the probability-weighted inputs and re-optimise under Yale-style caps:

$$\boldsymbol{\mu}_{\text{mix}} = \hat{\boldsymbol{\mu}} + \sum_s p_s \, \Delta\boldsymbol{\mu}_s, \qquad \max_{\boldsymbol{w}} \; \boldsymbol{w}^\top \boldsymbol{\mu}_{\text{mix}} - \tfrac{\lambda}{2} \boldsymbol{w}^\top \Sigma \boldsymbol{w}$$

`MeanRiskOptimizer` with `WeightBounds(0, 0.30)` mirrors the allocation limits Swensen imposes so the optimiser cannot short or concentrate.

Push it further

A crisis is not just lower $\boldsymbol{\mu}$ — correlations rise. Scale Σ per scenario and average; watch the allocation move as you re-weight the scenarios.

What the UBS case asks

You build your *own* risk-parity model and defend it:

- The 2018 / 2019 exposures and performance of your strategy
- How you define risk contributions and estimate the covariance
- Advantages and drawbacks vs. a balanced 60/40 portfolio
- Where you would invest more research effort

The notebook gives a working baseline — four clusters (equity, bond, credit, inflation), an EWMA covariance, and an equal-risk-contribution solve.

Clusters and risk contributions

Euler's theorem splits portfolio volatility into additive contributions:

$$\sigma(\boldsymbol{w}) = \sqrt{\boldsymbol{w}^\top \boldsymbol{\Sigma} \boldsymbol{w}} = \sum_i \underbrace{w_i \frac{(\boldsymbol{\Sigma}\boldsymbol{w})_i}{\sqrt{\boldsymbol{w}^\top \boldsymbol{\Sigma} \boldsymbol{w}}}}_{\mathrm{RC}_i}$$

Risk parity sets every RC_i equal — needing only $\boldsymbol{\Sigma}$, never $\boldsymbol{\mu}$. Two routes in the notebook: a from-scratch SLSQP objective and the repo's `RiskBudgetOptimizer` (Spinu log-barrier) — they agree.

Key insight

Capital is *not* risk: an equal-capital cluster mix loads most of its risk on equities. Risk parity tilts capital to bonds to balance the contributions.

Estimating the covariance

The only inputs are volatilities and correlations — far easier to estimate than μ. The paper uses rolling *daily* vol and *weekly* correlation windows; the notebook compares:

sample equal-weight over all history

EWMA recent observations weighted more (`halflife`)

```
compute_covariance(returns, "ewma", halflife=24) * 12
```

Annualise ($\times 12$) before reading volatilities or sizing leverage; the risk-parity *weights* are scale-invariant to Σ.

Volatility targeting and results

Lever the (low-vol) risk-parity portfolio to a target: leverage $= \sigma_{\text{target}}/\sigma(\boldsymbol{w})$. Net of costs, monthly-rebalanced, 8% target vs. 60/40:

	Risk parity (8%)	60/40
CAGR	8.4%	7.6%
Volatility	8.6%	9.1%
Sharpe	0.99	0.86
2018 / 2019	−3.9% / +20.7%	−6.9% / +14.1%

Diversified *risk* plus leverage beats the equity-dominated 60/40 on risk-adjusted return — the paper's central claim, reproduced.

What can go wrong

Proxies are not the portfolio

Public proxies for Yale's private programmes miss manager alpha, fees, and the true illiquidity — treat the backtest as a *lower bound* on the story.

- Unsmoothing with too small a sample, or an unstable $\rho \to 1$, blows up the variance — cap it with `max_beta`
- A 1-month covariance implies *absurd* leverage — burn in before the first risk-parity rebalance
- Leverage carries funding and gap risk; the bond-heavy mix is exposed to a rate shock — the table hides that tail
- Scenario shifts are *your* assumptions, not data — state them

Key takeaways

- One `Backtester` and `TimeseriesAnalyzer` drive both cases
- Always unsmooth illiquid returns before judging risk-adjusted performance
- Benchmark a Yale mix against a risk-matched ACWI/Agg split to isolate the allocation from manager selection
- Scenario tilts live in μ *and* Σ; probabilities are a lever, not a fact
- Risk parity needs only Σ; leverage turns a low-vol diversified core into a competitive return

Further reading

Yale case Lerner, Tango & Ferragamo, *Yale Investments Office: November 2020*, HBS 9-821-074.

UBS paper Koester et al., *Risk Parity Portfolio*, UBS CIO GWM, 2019.

Geltner (1993) Unsmoothing appraisal-based returns.

Roncalli (2013) The standard reference on risk parity and budgeting.

Spinu (2013) The log-barrier solver behind `RiskBudgetOptimizer`.

Notation Reference

Portfolio and asset notation

Portfolio quantities

$\boldsymbol{w}$	weights vector ($N \times 1$)
w_i	weight on asset i
$\boldsymbol{w}^*$	optimal weights
$\boldsymbol{w}_T$	tangency portfolio
μ_p	expected return
σ_p	volatility
S	Sharpe ratio

Asset-level inputs

$\boldsymbol{\mu}$	expected returns ($N \times 1$)
μ_i	return on asset i
$\boldsymbol{\Sigma}$	covariance matrix ($N \times N$)
σ_i	volatility of asset i
ρ_{ij}	correlation, assets i and j
r_f	risk-free rate
$\hat{\boldsymbol{\mu}}, \hat{\boldsymbol{\Sigma}}$	sample estimates

Operators, indices, and conventions

Operators

$\mathbb{E}[\cdot]$	expectation
$\mathrm{Var}(\cdot)$	variance
$\mathrm{Cov}(\cdot, \cdot)$	covariance
$\mathrm{Corr}(\cdot, \cdot)$	correlation
$\mathbf{1}$	vector of ones
$\cdot^\top$	transpose
$\mathbb{R}$	real numbers

Indices and dimensions

N	number of assets
T	time periods in sample
K	number of factors
i, j	asset indices
t	time index
k	factor index

Typography

Bold lowercase $\boldsymbol{w}$: vectors

Bold uppercase $\boldsymbol{\Sigma}$: matrices

Italic w_i, μ_i: scalars

Glossary

Glossary: A–F

Alpha Return unexplained by exposure to common risk factors; the intercept in a factor-model regression.

Asset allocation The discipline of dividing a portfolio across broad asset categories to meet an investor's return objective and risk tolerance.

Efficient frontier The set of portfolios offering the highest expected return for any given volatility.

Expected return The probability-weighted average future return of an asset or portfolio, denoted μ_i or μ_p; always estimated, never known.

Glossary: M–T

Mean-variance optimization (MVO) Markowitz's (1952) framework: construct portfolios by trading off expected return against variance. The efficient frontier and tangency portfolio are its key outputs.

Rebalancing Periodically restoring portfolio weights to target values after they drift due to differential returns.

Sharpe ratio Excess return per unit of volatility: $S = (\mu_p - r_f)/\sigma_p$. The standard single-number measure of risk-adjusted performance.

Tangency portfolio The risky portfolio that maximises the Sharpe ratio, denoted $\boldsymbol{w}_T$. Under the one-fund separation theorem, every mean-variance investor holds it combined with the risk-free asset.

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